

DMS201: Introduction to Management

Module-3: Operations Management

(Production Management)

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Introduction to Operations Management

(Issues affecting Supply Chain and its Operations)

- E-Commerce, e.g., Business to Business (B2B)/Business to Customer (B2C)/Customer to Customer (C2C) and increase of such ideas.
- High consumer expectation and demand.
- Globalization.
- Government rule/regulation.
- Infrastructure, Logistics, Transportation system.

Introduction to Operations Management

(Issues affecting Supply Chain and its Operations)

- Green Supply Chain Management (SCM) and reduction of wastage.
- Traditional seller/vendors/shops.
- Sourcing issue and how contracts can be formulated.
- Vertical/Horizontal integration of suppliers/vendors, etc., and comparison based on in housing production/manufacturing.

Introduction to Operations Management

(SC: Single v/s Multiple Suppliers/Vendors)

- Relationship between company and vendor(s).
- Contract duration between company and vendor(s).
- Purchase discount.
- Delivery scheduling and coordination.

Introduction to Operations Management (**SC: Single v/s Multiple Suppliers/Vendors**)

- Negotiation for purchase/order volume.
- Coordination.
- Application of Operations Management techniques like Just in Time (JIT), Total Quality Management (TQM), etc.

Introduction to Operations Management

(Benefit: Manufacturer v/s Supplier)

- Cost versus Volume of managing the operation.
- Inventory/Stock volume/cost.
- Ability/ease to approach market/customer.
- Loyalty of customer.
- Implementation of new manufacturing technique/method.
- Implementation of new labour rule/law.

Introduction to Operations Management

(Benefit: Manufacturer v/s Supplier)

- Uncertainty issues related to raw material, man power, supply, etc.
- Short versus Long term planning.
- Information sharing and ease and effectiveness of actions.
- Sharing of opportunity and risk.
- Synchronicity of long/short term mission/vision of market/business, etc.
- Ability to handle customer/other party issues.

Introduction to Operations Management

(SCM: Warehouse, Distribution Centres, etc.)

- Used for storing goods for short/long periods.
- The three important roles are: transportation consolidation, product mixing or blending, and service.
- **Crossdocking** is a concept which is used to eliminate storage and order-picking functions of a distribution warehouse while still performing its receiving and shipping functions. Vehicles arrive at a cross dock with goods to be sorted, consolidated with other products, and loaded onto transport vehicles.
- Types of cross docking: Manufacturing crossdocking, Distributor crossdocking and Retail crossdocking

Production Management

- Introduction to production management
- **Capacity planning and process analysis**
- Facility location
- Aggregate production planning
- Materials requirement planning
- Sequencing and scheduling

Capacity planning and process analysis

Capacity planning and process analysis

- For a rationale production planning based on a set capacity plan (or to be decided plan) three basic issues must be resolved and these are:
 - ❖ How often the inventory status should be determined
 - ❖ When a replenishment order should be placed
 - ❖ How large the replenishment order should be

Capacity planning and process analysis (Inventory/stock)

- Six broad decision categories for controlling aggregate inventory/stock are:
 - ❖ Cycle inventory/stock
 - ❖ Congestion inventory/stock
 - ❖ Safety (or buffer) inventory/stock
 - ❖ Anticipation inventory/stock
 - ❖ Pipeline inventory/stock
 - ❖ Decoupling inventory/stock

Capacity planning and process analysis (Cycle inventory/stock)

- Produced or ordered in batches instead of one unit at a time and the amount of inventory/stock in hand, at any point, that results from these batches is called cycle stock.
- Reason for a batch rather than one unit is to do with economies of scale, quantity discounts, technological restrictions.
- The amount of cycle inventory/stock in hand at any time depends directly on how frequently orders are placed.

Capacity planning and process analysis (Congestion inventory/stock)

- Results due to items competing for limited capacity.
- When multiple items share the same production equipment, particularly when there are significant setup times, inventories of these items build up as they wait for the equipment to become available.

Capacity planning and process analysis

(Safety inventory/stock)

- Is the amount of inventory/stock kept on hand, on the average, to allow for the uncertainty of demand and the uncertainty of supply in the short run.
- It is not needed when the future rate of demand and the length of time it takes to get complete delivery of an order are known with certainty.
- Thus the level of safety inventory/stock is controllable in the sense that this investment is directly related to the desired level of customer service (i.e., how often customer demand is met from the inventory/stock).

Capacity planning and process analysis (**Anticipation inventory/stock**)

- It is that inventory/stock which is accumulated in advance in anticipation of an expected peak in sales.
- When demand is regularly lower than average during some parts of the year, excess inventory (above cycle and safety stock) can be built up so that, during the period of high anticipated requirements, extra demand can be serviced from stock rather than from, for example, working overtime in the plant.
- Thus aggregate production rate can be stabilized through the use of anticipatory inventory/stock.

Capacity planning and process analysis (Anticipation inventory/stock)

- Anticipation inventory/stock can also occur because of seasonality of supply, anticipation of price change of raw materials, scarcity of raw materials, etc.
- Anticipation inventory/stock can also consist of inventory/stock built up to deal with labour strikes, war crises, or any other events that are expected to result in a period of time during which the rate of supply is likely to be lower than the rate of demand.

Capacity planning and process analysis **(Pipeline inventory/stock or Work in Progress (WIP))**

- It includes goods in transit, between levels of a multi-echelon distribution system or between adjacent work stations in a factory.
- The pipeline inventory/stock of an item between two adjacent locations is proportional to the usage rate of the item and to the transit time between the locations.

Capacity planning and process analysis (Decoupling inventory/stock)

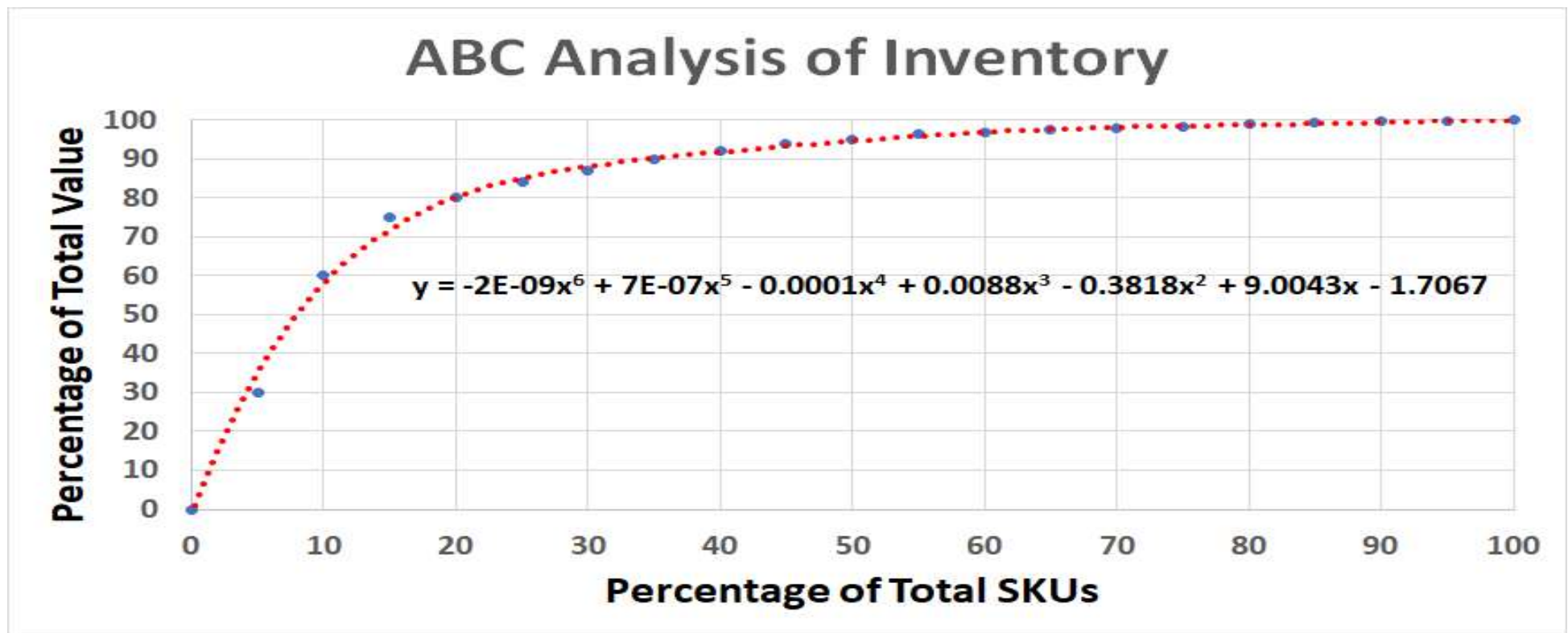
- It is used in a multi-echelon situation to permit the separation of decision making at the different echelons.
- Decoupling inventory/stock allows decentralized decision making at branch warehouses without every decision at a branch having an immediate impact.
- Before taking a look at decoupling inventory/stock one should understand that this inventory/stock level can impact the demand of customers as well as that of the down stream production planning sequence.

Capacity planning and process analysis

(ABC Analysis of Inventory)

- **Stock-Keeping Unit (SKU)**: It is the specific unit of stock to be controlled and is defined as an item of stock that is completely specified as to function, style, size, color, and, often, location.
- P.K.Kelkar library, IIT Kanpur (26.5123° N, 80.2333° E) can have same course (DMS201) books assigned as SKU, while Springer can have books for DMS201 with different ISBN assigned as SKU.
- Typically, 20% of the SKUs account for 80% of the total value, 30% of the SKUs account for 15% of the total value and 50% of the SKUs account for 5% of the total value.
- Based on % of SKUs and % of total value we have the **Distribution by Value (DBV)** curve, also called **ABC Analysis of inventory**.

Capacity planning and process analysis (ABC Analysis of Inventory)



Capacity planning and process analysis (ABC Analysis of Inventory: Log-Normal Distribution)

$$f(x) = \{1/x\sigma\sqrt{(2\pi)}\} \exp\{-(\log_e x - \mu)^2/2\sigma^2\}; 0 < x < \infty$$

- μ (logarithmic of location), σ^2 (σ = logarithmic of scale) are the parameters where $\mu \in \mathbb{R}$ and $\sigma^2 > 0$
- $E(X) = \mu_m = \exp\{\mu + (\sigma^2/2)\}$
- $V[X] = \{\exp(\sigma^2) - 1\} \exp(2\mu + \sigma^2)$
- Median = $\mu_e = \exp(\mu)$
- Mode = $\mu_o = \exp(\mu - \sigma^2)$
- Example: Stock prices return distribution, ABC Analysis of Inventory

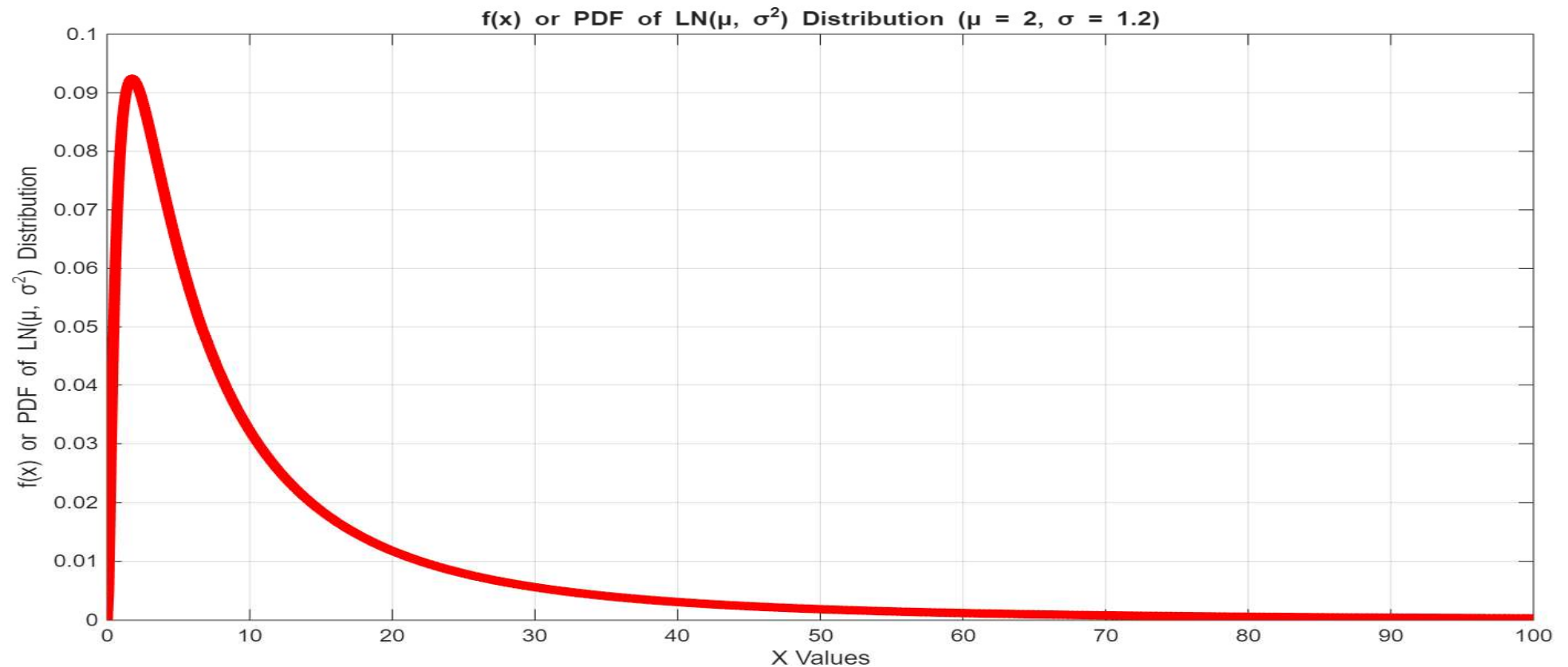
Capacity planning and process analysis

(ABC Analysis of Inventory: Log-Normal Distribution)

```
1. %PLOT OF PDF OF LOG NORMAL DISTRIBUTION LN( $\mu$ ,  $\sigma^2$ )
2. % Define the range of x values for the plot
3. a = 0; % Minimum value of X as needed to plot
4. b = +100; % Some maximum value of X as needed to plot
5. mu = 2.0; % Location parameter
6. sigma = 1.2; % Scale parameter
7. % Define the range of X values for the plot
8. x = a:0.1:b;
9. % Calculate the PDF values using lognpdf
10. y = lognpdf(x, mu, sigma);
11. % Plot the PDF of LN( $\mu$ ,  $\sigma^2$ )
12. plot(x, y, 'LineWidth', 5); % Plot x vs y with a specified line width;
13. % Add labels and title for clarity
14. xlabel('X Values'); % Label the x-axis
15. ylabel('f(x) or PDF of LN( $\mu$ ,  $\sigma^2$ ) Distribution'); % Label the y-axis
16. title(['f(x) or PDF of LN( $\mu$ ,  $\sigma^2$ ) Distribution ( $\mu$  = ', num2str(mu), ',  $\sigma$  = ', num2str(sigma), ')]); % Set the plot title
17. % Optional: Add a grid for better readability
18. grid on; % Add a grid to the plot
```

Capacity planning and process analysis

(ABC Analysis of Inventory: Log-Normal Distribution)



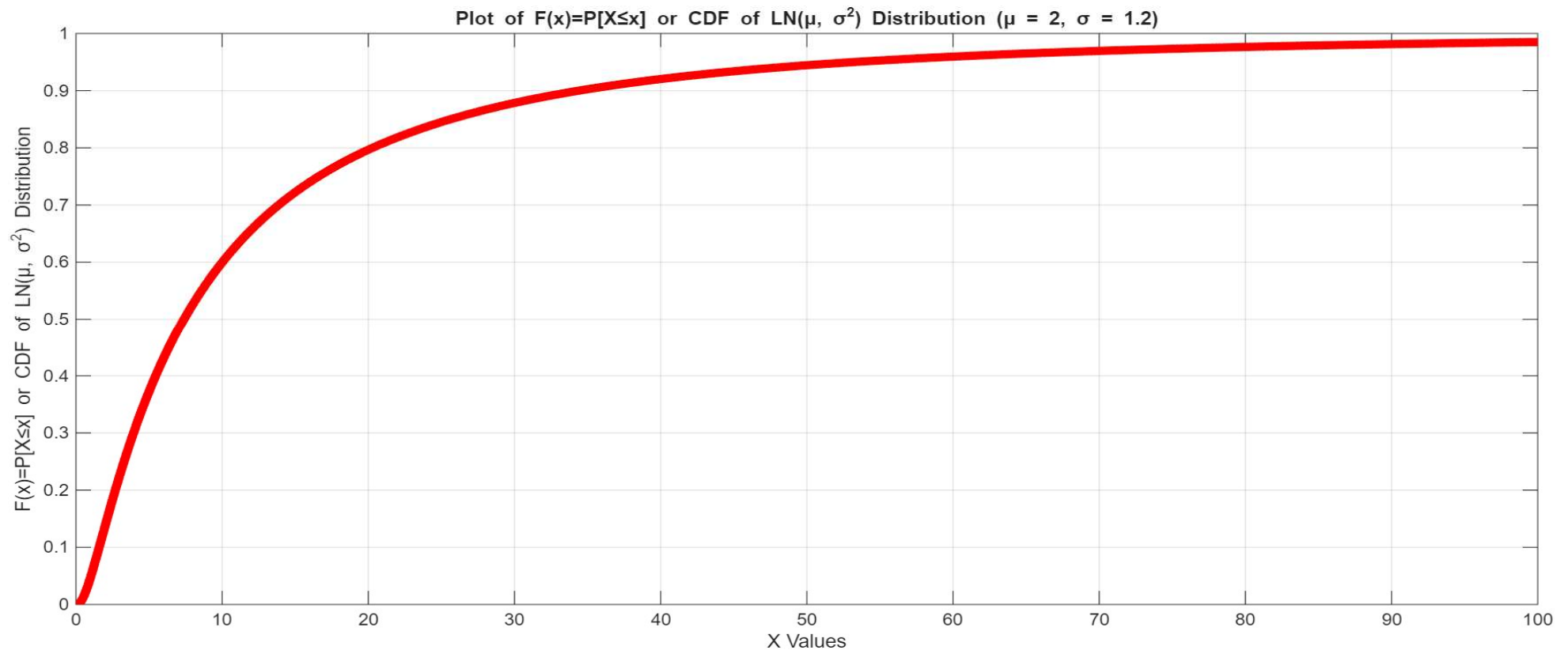
Capacity planning and process analysis

(ABC Analysis of Inventory: Log-Normal Distribution)

```
1. %PLOT OF CDF OF LOG NORMAL DISTRIBUTION LN( $\mu$ ,  $\sigma^2$ )
2. % Define the range of X values for the plot
3. a = 0; % Minimum value of X as needed to plot
4. b = +100; % Some maximum value of X as needed to plot
5. mu = 2.0; % Location parameter
6. sigma = 1.2; % Scale parameter
7. % Define the range of X values for the plot
8. x = a:0.1:b;
9. % Calculate the PDF values using logncdf
10. y = logncdf(x, mu, sigma);
11. % Plot the CDF of LN( $\mu$ ,  $\sigma^2$ )
12. plot(x, y, 'LineWidth', 5); % Plot x vs y with a specified line width;
13. % Add labels and title for clarity
14. xlabel('X Values'); % Label the x-axis
15. ylabel('F(x)=P[X≤x] or CDF of LN( $\mu$ ,  $\sigma^2$ ) Distribution'); % Label the y-axis
16. title(['Plot of F(x)=P[X≤x] or CDF of LN( $\mu$ ,  $\sigma^2$ ) Distribution ( $\mu$  = ', num2str(mu), ',  $\sigma$  = ', num2str(sigma), ')]); % Set the plot title
17. % Optional: Add a grid for better readability
18. grid on; % Add a grid to the plot
```

Capacity planning and process analysis

(ABC Analysis of Inventory: Log-Normal Distribution)



Capacity planning and process analysis

(ABC Analysis of Inventory : Example # 13)

SNo	Item/Product	Unit Cost (INR)	Annual Usage (Units)	Annual Value (INR)
01	Product # 01	200	17,500	$200 \times 17500 = 35,00,000$
02	Product # 02	30	1,00,000	$30 \times 100000 = 30,00,000$
03	Product # 03	900	1,500	$900 \times 1500 = 13,50,000$
04	Product # 04	150	500	$150 \times 500 = 75,000$
05	Product # 05	500	2,000	$500 \times 2000 = 10,00,000$
06	Product # 06	450	800	$450 \times 800 = 3,60,000$
07	Product # 07	300	3,000	$300 \times 3000 = 9,00,000$
08	Product # 08	770	3,500	$770 \times 3500 = 26,95,000$
09	Product # 09	15	2,6,000	$15 \times 26000 = 3,90,000$
10	Product # 10	7,500	155	$7500 \times 155 = 11,62,500$
Total				1,44,32,500

Capacity planning and process analysis

(ABC Analysis of Inventory : Example # 13)

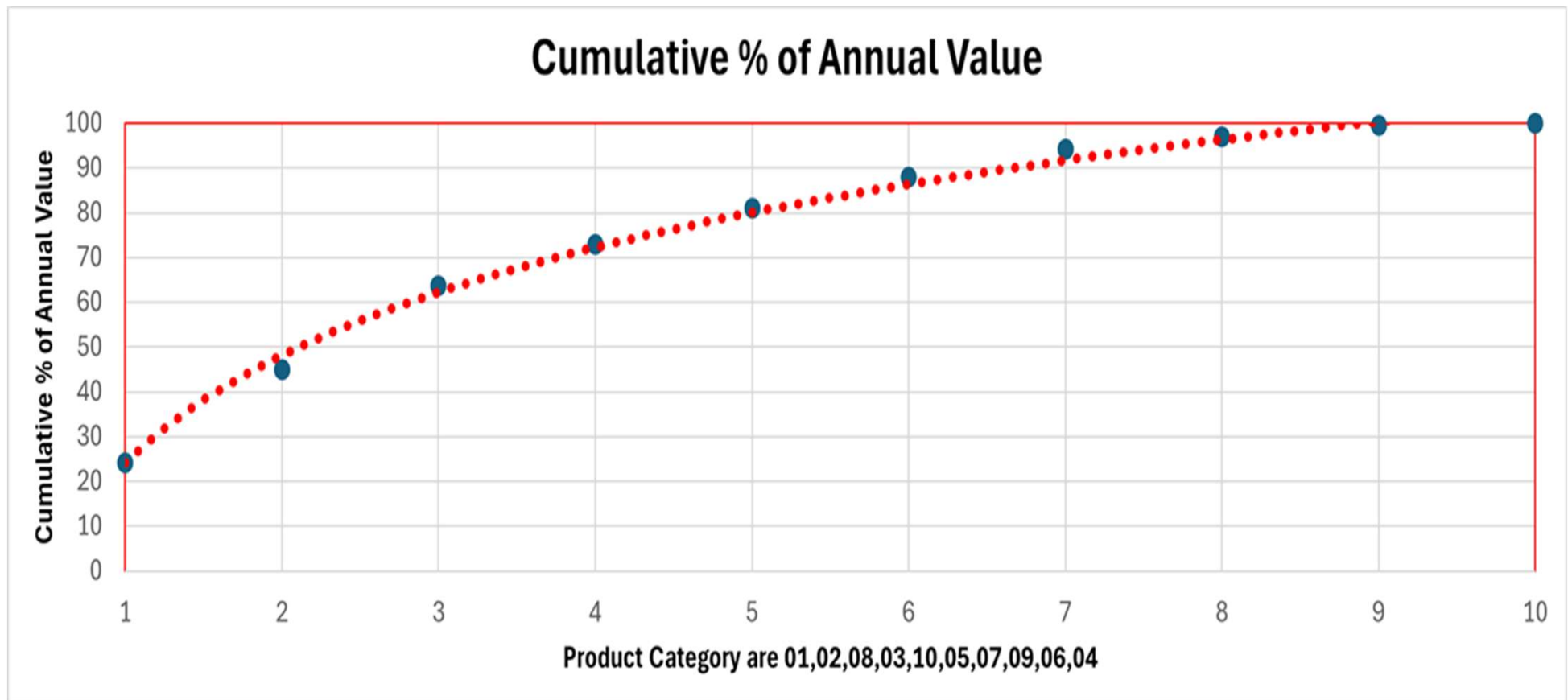
SNo	Item/Product	Unit Cost (INR)	Annual Usage (Units)	Annual Value (INR)
01	Product # 01	200	17,500	35,00,000
02	Product # 02	30	1,00,000	30,00,000
08	Product # 08	770	3,500	26,95,000
03	Product # 03	900	1,500	13,50,000
10	Product # 10	7,500	155	11,62,500
05	Product # 05	500	2,000	10,00,000
07	Product # 07	300	3,000	9,00,000
09	Product # 09	15	2,60,00	3,90,000
06	Product # 06	450	800	3,60,000
04	Product # 04	150	500	75,000
	Total			1,44,32,500

Capacity planning and process analysis

(ABC Analysis of Inventory : Example # 13)

SNo	Item/Product	Annual Value (INR)	% of Annual Value	Cumulative % of Annual Value
01	Product # 01	35,00,000	24.25	24.25
02	Product # 02	30,00,000	20.79	45.04
08	Product # 08	26,95,000	18.67	63.71
03	Product # 03	13,50,000	9.35	73.06
10	Product # 10	11,62,500	8.05	81.12
05	Product # 05	10,00,000	6.93	88.05
07	Product # 07	9,00,000	6.24	94.28
09	Product # 09	3,90,000	2.70	96.99
06	Product # 06	3,60,000	2.49	99.48
04	Product # 04	75,000	0.52	100.00
	Total	1,44,32,500	100	

Capacity planning and process analysis (ABC Analysis of Inventory : Example # 13)



Capacity planning and process analysis

- **Capacity planning and process analysis** affects all areas of an operation, and it measures the operation can transform inputs into outputs.
- The concept is about the quantity of a product or service that can be made within a given time period. For example, the number of passengers per flight on an INDIGO aeroplane, the number of patients that can be seen by the OPD at HC, IIT Kanpur, the number of mobile phones that can be produced in a week by SAMSUNG.
- It is defined as the number of units (goods and/or services) an operation can produce over a given time period, under normal working conditions, where no additional resources are deployed.
- Capacity is usually measured in convenient units such as litres per hour or passengers per taxi, e.g., a domestic tap may be able to deliver 20 litres per minute of water; a bus may have a capacity of 53 passengers, a football stadium may be able to seat 50,000 spectators or Hall 8 mess at IIT Kanpur may be able to serve 275 students per hour.

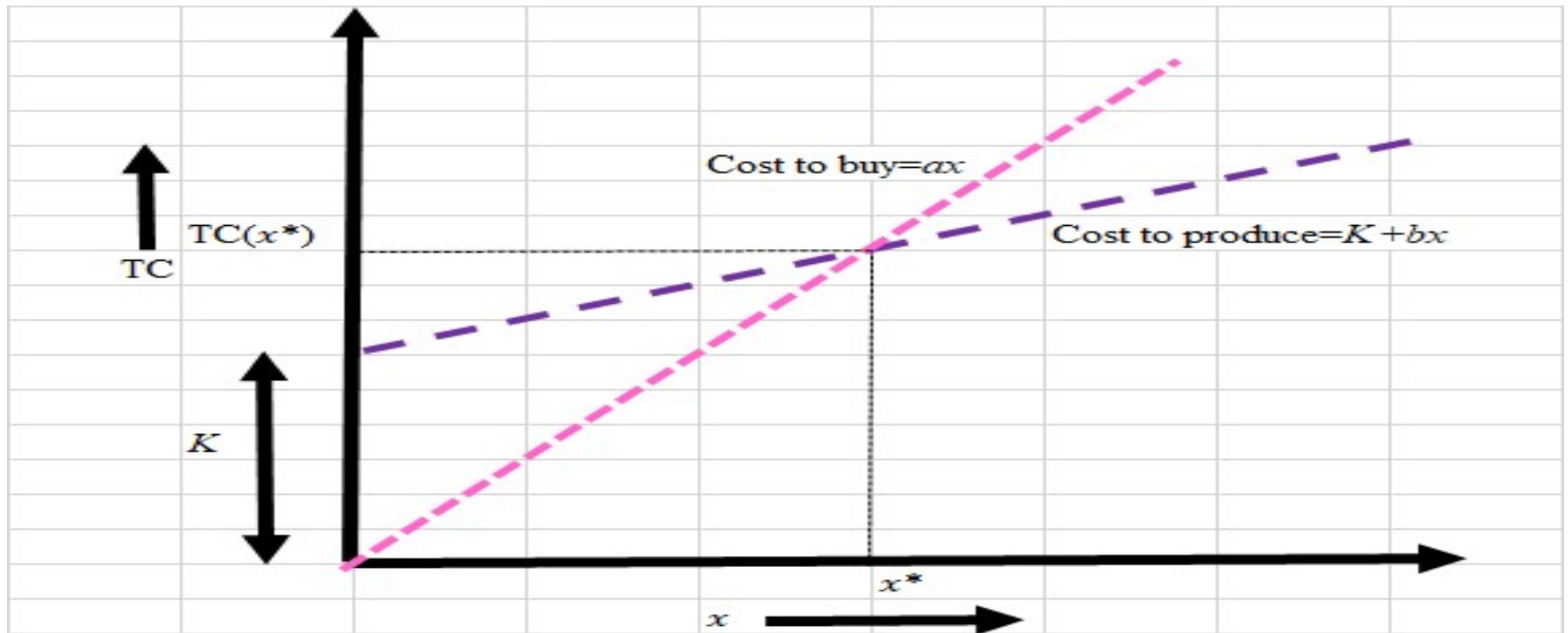
Capacity planning and process analysis

- A classic problem faced by the firm is known as the make-or-buy decision.
- The firm can purchase the product from an outside source for a per unit, but can produce it internally for a lower unit price b such that $b < a$.
- However, in order to produce the product internally, the company must invest Rs. K to expand production capacity.
- Which strategy should the firm adopt

Capacity planning and process analysis

- The total cost of the firm to produce x units is $(K + bx)$, which is equivalent to $(K/x + b)$ per unit.
- As x increases, the cost per unit of production decreases, since (K/x) is a decreasing function of x .
- The cost to purchase outside is a per unit, independent of the quantity ordered (provided there are no discounts).
- By graphing the total costs of both internal production and external purchasing, we can find the point at which the costs are equal.
- This is known as the **break-even quantity** $x = \{K/(a - b)\}$

Capacity planning and process analysis



Capacity planning and process analysis (**Example # 14**)

- Kirloskar Electric is designing a new model of vacuum circuit breakers and must decide whether to produce the casing of the electronic circuit internally or to purchase them from an outside supplier.
- The supplier is willing to produce the casing and sell it for Rs. 2,200 each, but the manufacturer estimates that the firm can produce the keyboards for Rs. 1,750/- each.
- Management estimates that expanding the current plant and purchasing the necessary equipment to make the casing of the electronic circuit would cost Rs. 80,00,000.
- Should they undertake the expansion?
- The **break even quantity** is $8000000 / (2200 - 1750) \approx 17778$ units, so the manufacturer has to sell a minimum of 17,778 units of circuit breakers to justify the investment to product the casing of the electronic circuit for the vacuum circuit breakers

Capacity planning and process analysis (with discounts)

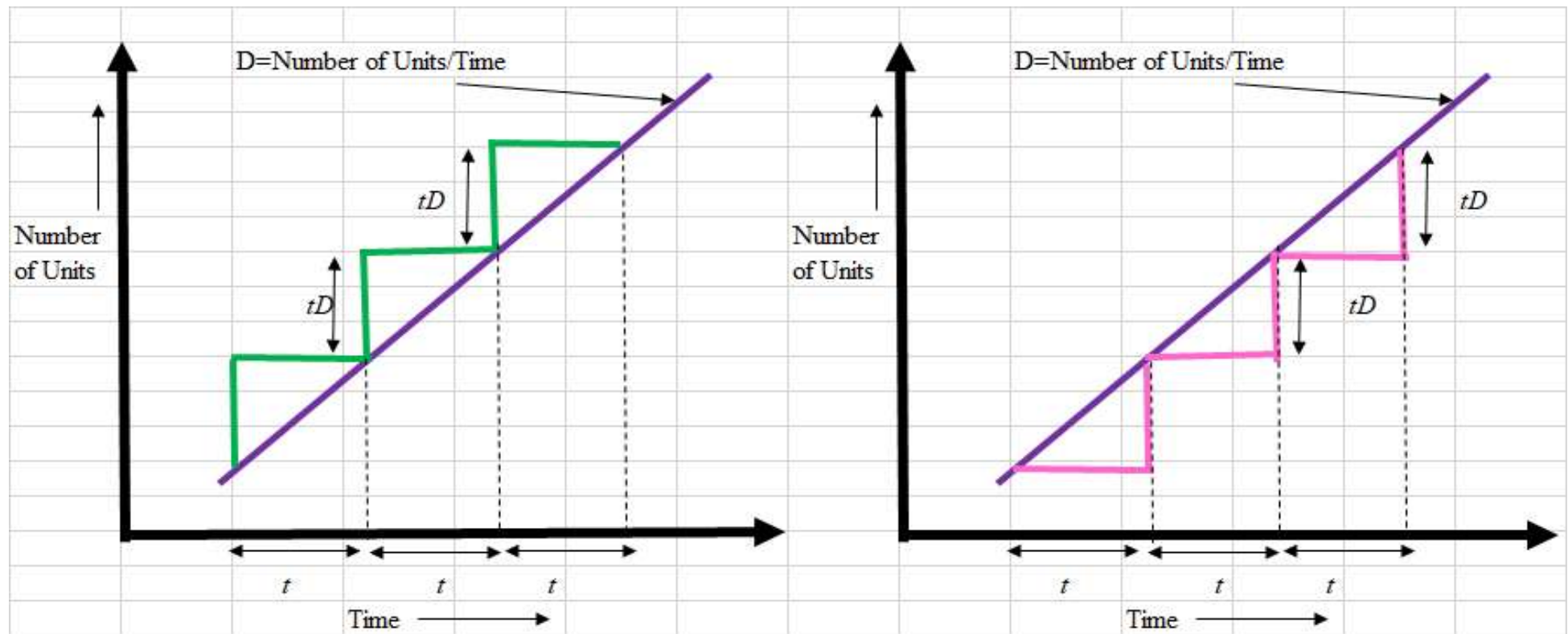


Capacity planning and process analysis

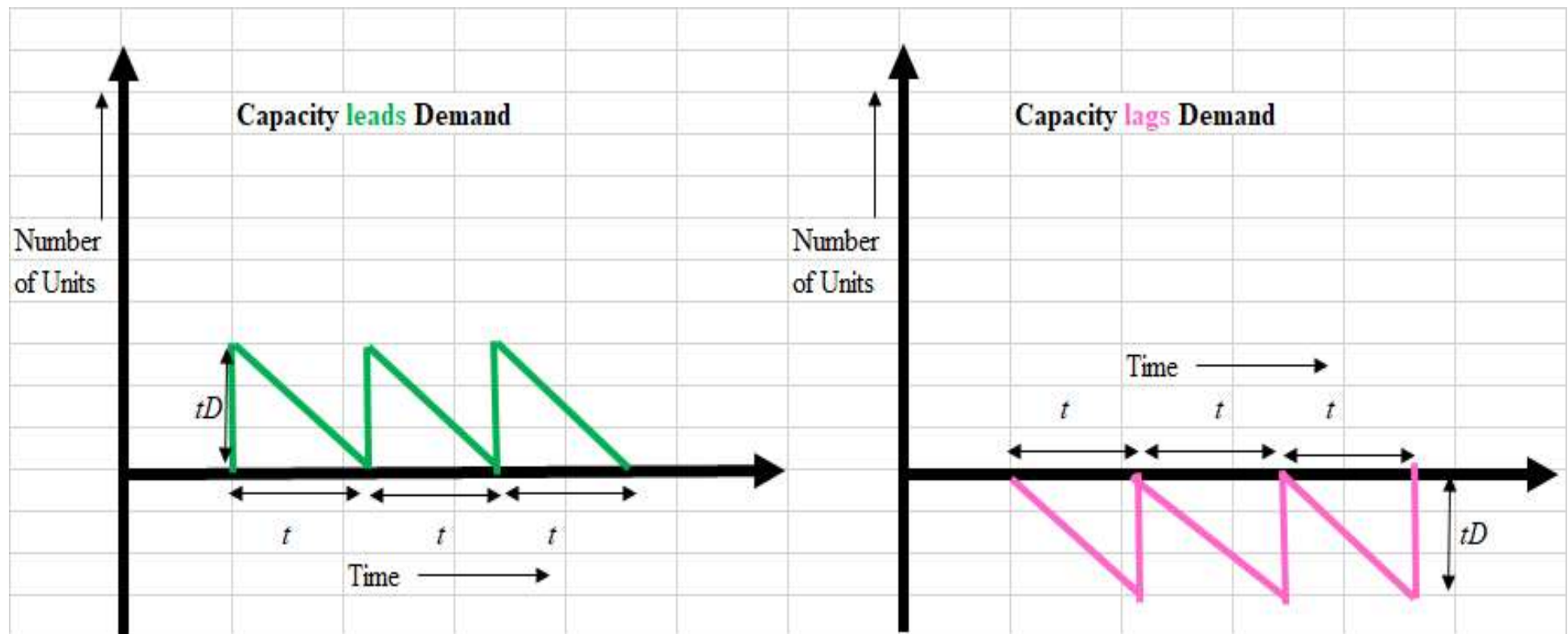
- Under **Dynamic Capacity Expansion Policy**, capacity decisions must be made based on dynamics of the changing demand pattern.
- Two competing objectives in capacity planning are
 - ❖ **Maximizing market share**: Increase it to capture the market. When customer demand suddenly falls then the firm has to handle inventory piles up. Other problem can happen based on changed demand preference of customers. Furthermore, for price sensitive customer demand it may be difficult for the firm to change the price even it warranted.
 - ❖ **Maximizing capacity utilization**: When demand is high than capacity the firm faces risk of incurring shortages, else inventory piles up if demand is on the lower side. Other problems is changing requirement of customer and inventory can be obsolete. With some capacity cushion excess capacity can be used to respond to sudden demand surges and it puts the firm in a position to capture a larger portion of the market place if the opportunity arises.

Capacity planning and process analysis

(Capacity leads/lags demand)



Capacity planning and process analysis (Capacity leads/lags demand)



Capacity planning and process analysis (**Capacity Planning and Expansion**)

- **Finite plant lifetime:** Plants/units have finite lifetime, and companies do close plants for a variety of reasons, like business shut down (***Kodak: Shut down their business***), equipment becomes obsolete or unreliable and cannot be replaced easily (***Nokia: Sold their phone business***), labor costs or requirements too prohibitive (***Ford: Closed their units in India***), major changes in the process technology (***Timken: Closed their unit in Jamshedpur (22.8046° N, 86.2029° E)***), etc.

Capacity planning and process analysis

(Capacity Planning and Expansion)

- **Demand patterns:** Demand does not grow at a constant rate per year, and growth patterns can depend on uncertainty (*fall of real estate prices during negative growth in economy*), economic changes (*sales of vehicles in India after 7th pay commission*), climatic impact (*air purifier requirement in many cities in India*), technology change (*OTT platforms effecting DVDs and CDs market of movies*), unforeseen uncertainty (*demand of COVID-19 vaccines during 2020-22*), etc.

Capacity planning and process analysis (**Capacity Planning and Expansion**)

- **Technological developments:** Capacity enhancement costs are not constant and changes (*Maruti Suzuki utilizing demand forecasting techniques to optimize assembly line and manage component supply chain*).

Capacity planning and process analysis (**Capacity Planning and Expansion**)

- **Government regulation:** Environmental and government regulation regulate business/company (*ITC faces significant regulation due to evolving government policies, strict guidelines concerning the tobacco industry*).

Capacity planning and process analysis (Capacity Planning and Expansion)

- **Overhead costs:** Difference of cost related to capacity expansion and location models have effect on capacity planning (*Hindustan Motors was severely affected due to huge overhead and resistance to change by workers*).

Capacity planning and process analysis (Capacity Planning and Expansion)

- **Tax incentives:** Different countries/region/states give tax sops for new technology to facilitate increase in demand (*Ultratech Cement benefited by the decrease in GST*).

Capacity planning and process analysis

- The key questions in **capacity planning and process analysis** are:
 - ❖ What kind of capacity is needed?
 - ❖ How much is needed to match demand?
 - ❖ When is it needed?
 - ❖ How much will it cost?
 - ❖ What are the potential advantages/disadvantages?
 - ❖ Is the plan sustainable in the short/long run?
 - ❖ Can capacity changed be planned or is it fixed once designed and implemented?
 - ❖ Is a change in SCM, Vendor, Supplier, etc., needed as capacity changes?

Capacity planning and process analysis

(Effects of Capacity Planning and Expansion)

- Ability of the organization to meet future demands for products and services.
- Operating costs.
- Initial cost for set-up and starting production/services.
- Both short-term as well as long-term commitment of resources/manpower, etc., but in different proportions.
- Competition and market response.
- Ability to operate in local versus global markets.

Capacity planning and process analysis

- **Design capacity:** The maximum designed service capacity or output rate.
- **Effective capacity:** It is defined as **design capacity** minus allowances such as personal time, equipment maintenance, delays due to scheduling problems, and changing the mix of products.

Capacity planning and process analysis (**Example # 15**)

- **Efficiency** = (Actual output/Effective capacity)×100%.
- **Utilization** = (Actual output/Design capacity)×100%.
- Consider: Design capacity 50 trucks per day, Effective capacity 40 trucks per day, Actual output 36 trucks per day.
- Efficiency = $(36/40) \times 100 = 90\%$.
- Utilization = $(36/50) \times 100 = 72\%$.

Capacity planning and process analysis

(Factors affecting Capacity Planning)

- **Facilities:** Design, Location, Layout, Environment.
- **Product/service:** Design, Product or service mix.
- **Process:** Quantity, Quality.
- **Human factors:** Job content, Job design, Training and experience, Motivation, Compensation, Learning rates, Absenteeism and labour turnover.
- **Policy:** Company, Subsidiary.
- **Operational:** Scheduling, Materials management, Quality assurance, Maintenance policies, Equipment breakdowns
- **Supply chain:** Vendor, Supplier.
- **External factors:** Product standards, Safety regulations, Government regulations, International market tariff, etc., Unions, Pollution control standards

Capacity planning and process analysis

(Factors deciding Capacity Planning)

- An organization decides its capacity strategy based on
 - ❖ Short-term/Long-term demand patterns.
 - ❖ Technological changes/Innovation.
 - ❖ Strategy and behaviour of competitors.
 - ❖ Growth rate and variability of demand.
 - ❖ Fixed/Operational costs.
 - ❖ Availability of capital and other inputs (land, nature, mineral resources, etc.).
 - ❖ Changing rule/regulation, etc.

Capacity planning and process analysis

(Steps in Capacity Planning)

- 1) Estimate future capacity requirements.
- 2) Evaluate existing capacity/facilities and identify gaps.
- 3) Identify alternatives for meeting requirements.
- 4) Conduct financial analyses of each alternative.
- 5) Assess key qualitative issues for each alternative.
- 6) Select the alternative that will be best in the long term.
- 7) Implement the selected alternative.
- 8) Monitor results.

Capacity planning and process analysis

(Strategic Tools for Effective Capacity Planning)

- 1) Brain storming.
- 2) Buzz/Small group feedback.
- 3) Step-Ladder technique.
- 4) Strength Weakness Opportunity Threat (SWOT) Analysis.
- 5) Diagrammatical/Writing/Stick-pad memo Analysis.
- 6) What, Where, Why, When Analysis.
- 7) Mass Collaboration.
- 8) Thinking the Negative Analysis.

Capacity planning and process analysis

(Forecasting Capacity Planning/Requirement)

- Capacity planning decisions involve both **long-term** and **short-term** considerations.
- **Long-term** considerations relate to overall level of capacity, such as facility size, future competition, technology development, new products, new type of production, new materials/service concepts/ideas.
- **Short-term** considerations relate to probable variations in capacity requirements created by such things as seasonal, random, and irregular fluctuations in demand.

Capacity planning and process analysis (Forecasting Techniques for Capacity Planning)

- Simple Moving Average
- Adjusted Moving Average
- Exponential Moving Average
- Non-Adaptive Parameter Exponential Moving Average
- Adaptive Parameter Exponential Moving Average
- Trend Adjusted Forecasting Method
- Trend, Seasonality Adjusted Forecasting Method
- Trend, Seasonality, Cyclicity Adjusted Forecasting Method

Capacity planning and process analysis

(Example # 16)

- AMPL (Mahindra) in the city of Vijawada (16.5062° N, 80.6480° E) has done a detailed study and has the following data for spark plug it manufactures per month. The manufacturing department works one 8-hour shift, 25 days a month.

Month	Demand	Processing time (minutes) per unit	Month	Demand	Processing time (minutes) per unit
Jan	4000	5.0	July	3700	5.0
Feb	3500	5.5	Aug	3800	5.1
Mar	3200	5.2	Sep	4150	5.3
Apr	4100	5.3	Oct	3950	5.2
May	4050	4.9	Nov	3900	5.0
June	4200	4.8	Dec	3850	4.9

Capacity planning and process analysis

(Example # 16)

Month	Total processing time (minutes)	Month	Total processing time (minutes)
Jan	$4000 \times 5.0 = 20000$	July	$3700 \times 5.0 = 18500$
Feb	$3500 \times 5.5 = 19250$	Aug	$3800 \times 5.1 = 19380$
Mar	$3200 \times 5.2 = 16640$	Sep	$4150 \times 5.3 = 21995$
Apr	$4100 \times 5.3 = 21730$	Oct	$3950 \times 5.2 = 20540$
May	$4050 \times 4.9 = 19845$	Nov	$3900 \times 5.0 = 19500$
June	$4200 \times 4.8 = 20160$	Dec	$3850 \times 4.9 = 18865$

Capacity planning and process analysis

(Example # 16)

- Utilization = Processing time/Processing time capacity.
- Utilization = $(236405/60)/(8 \times 25) = 19.7 \approx 18$ machines are needed.
- Utilization = $\{(236405/60)/(8 \times 25)\}/0.95 = 20.7 \approx 21$ machines are needed if we consider 95% utilization of the machines per month.

Capacity planning and process analysis

(Key points for Capacity Planning)

- Exclusive focus on sales and revenue potential.
- Not taking care of optimal **product mix** for an optimal sales and revenues.
- Not a proper/scientific use of forecasting techniques to determine capacity requirements
- Too much emphasis on **short-term** demand/capacity planning rather than **long-term** demand/capacity planning.
- Optimal balance of vendor/supplier (taking into consideration their respective capacity/quality, etc.) versus in-house capability.

Capacity planning and process analysis

(Key points for Capacity Planning)

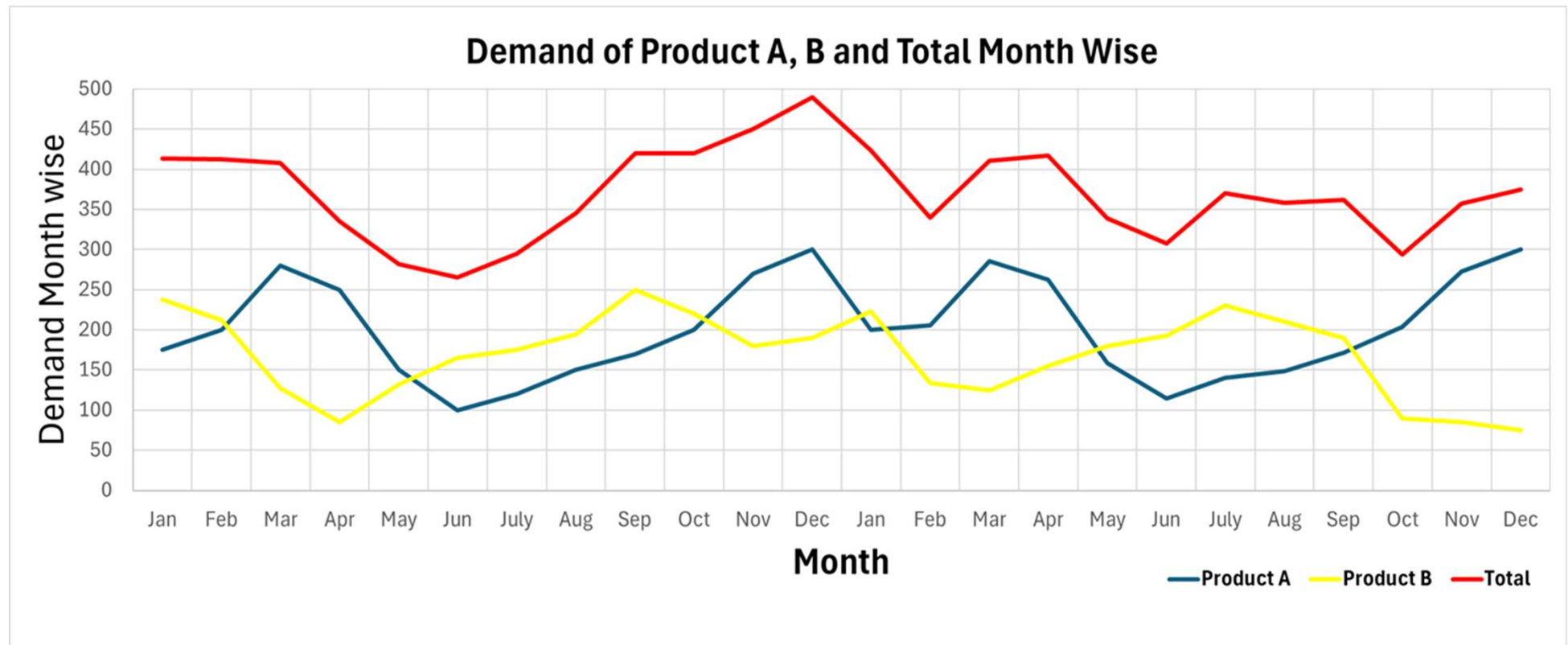
- Lack of expertise.
- Poor emphasis on **quality** versus **quantity**.
- Improper analysis of demand from customers.
- Ineffective cost consideration.
- Inability to consider all risk factors however negligible they are.

Capacity planning and process analysis

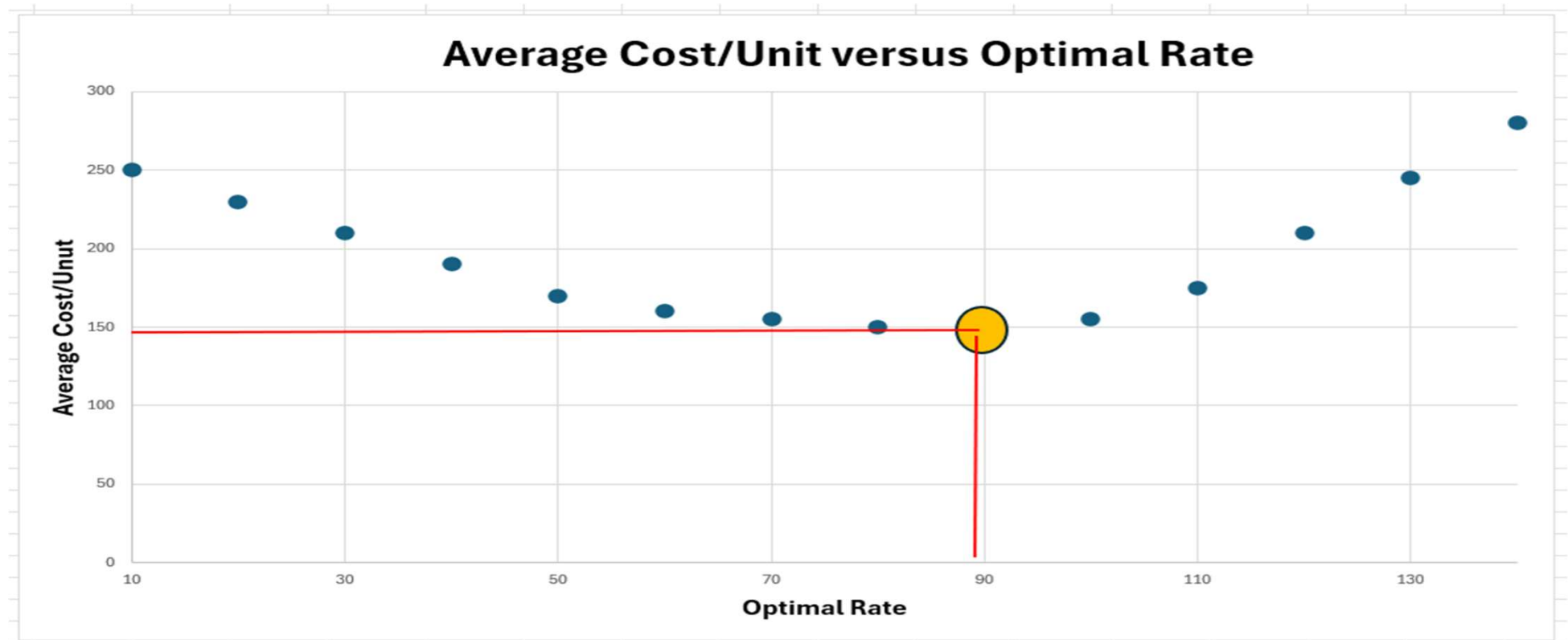
(Key points for Capacity Planning)

- Design flexibility in the capacity planning and production system
- Consider the **life** and **stage** of the product.
- Study **bottleneck** properly and analyze capacity build up in quantum depending on demand, manufacturing cost, manufacturing time, set-up time, in-between design time, etc.
- Smoothing of capacity planning is necessary and essential.
- Identification of optimal operational level (**economy of scale**) should always be considered.

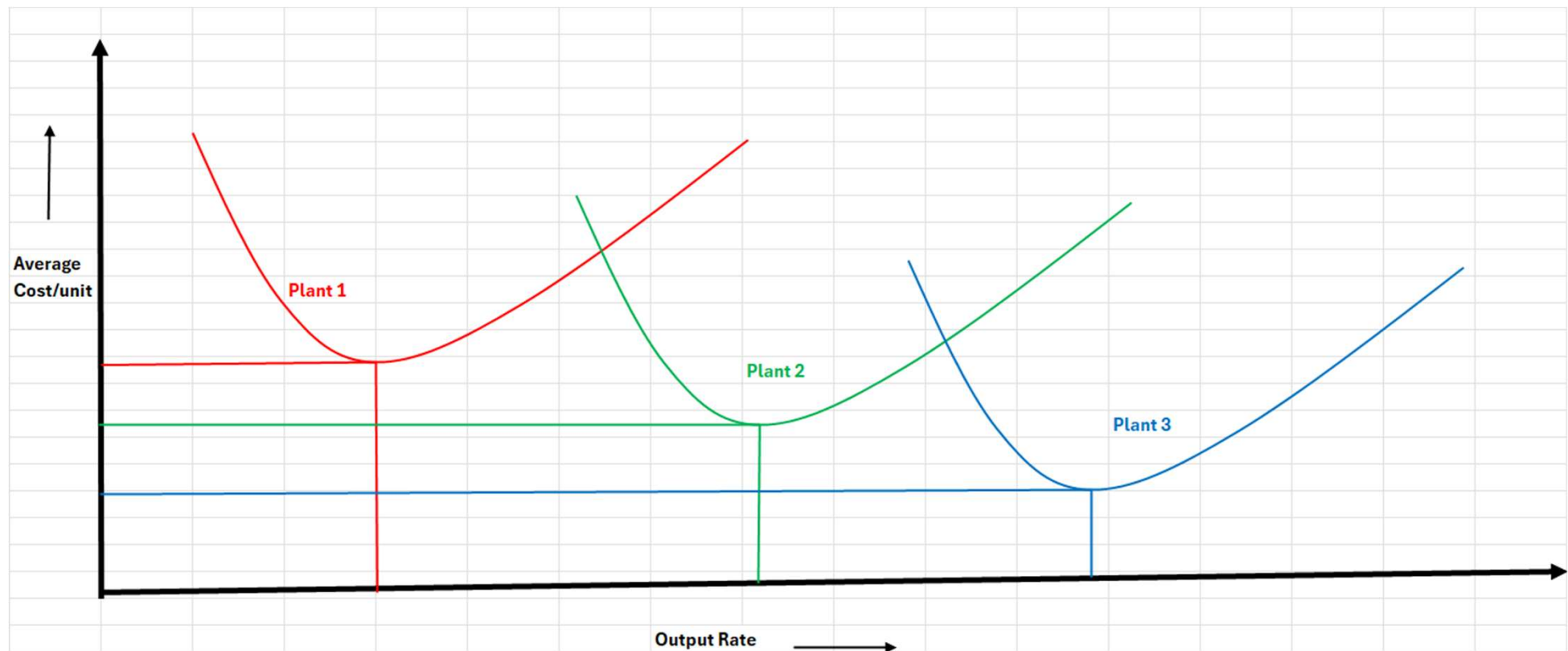
Capacity planning and process analysis



Capacity planning and process analysis (Economic Order Quantity (EOQ))



Capacity planning and process analysis



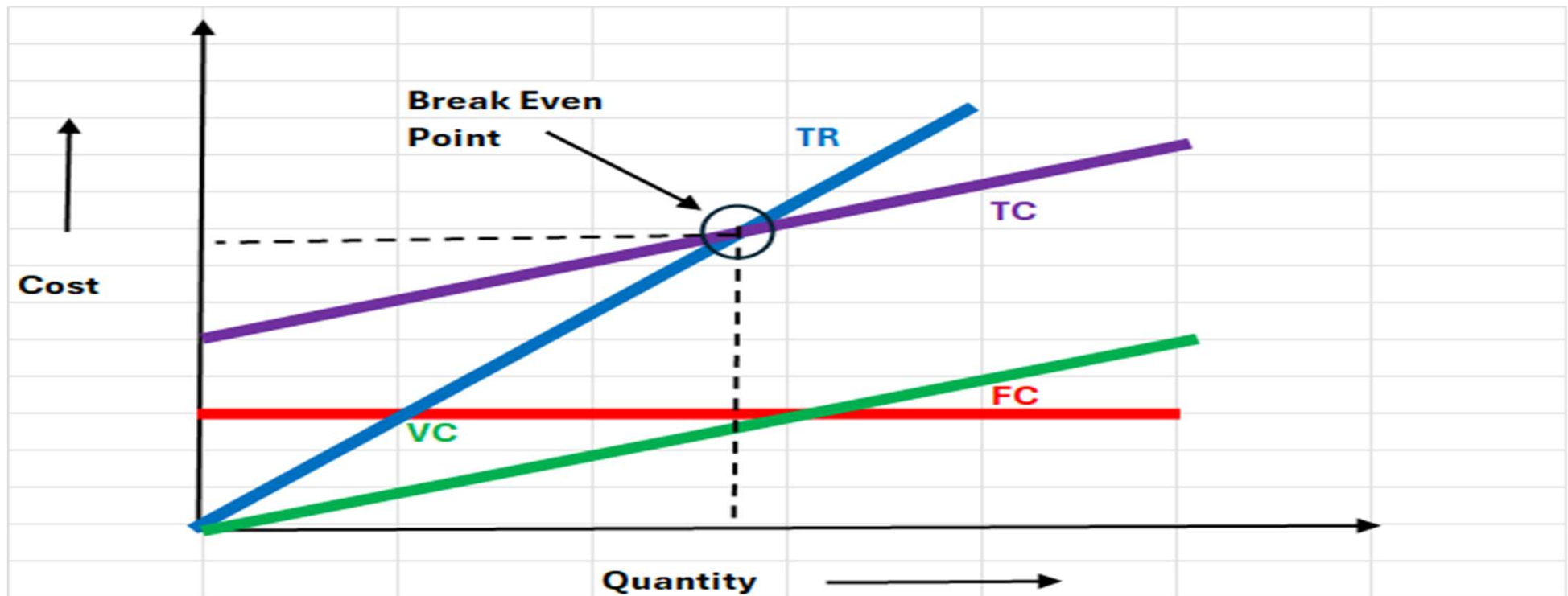
Capacity planning and process analysis

- FC = Fixed Cost
- VC = Total Variable Cost
- v = Variable Cost/Unit
- TC = Total Cost
- TR = Total Revenue
- R = Revenue per Unit
- Q = Quantity or Volume of Output
- Q_{BEP} = Break-Even Quantity
- P = Profit

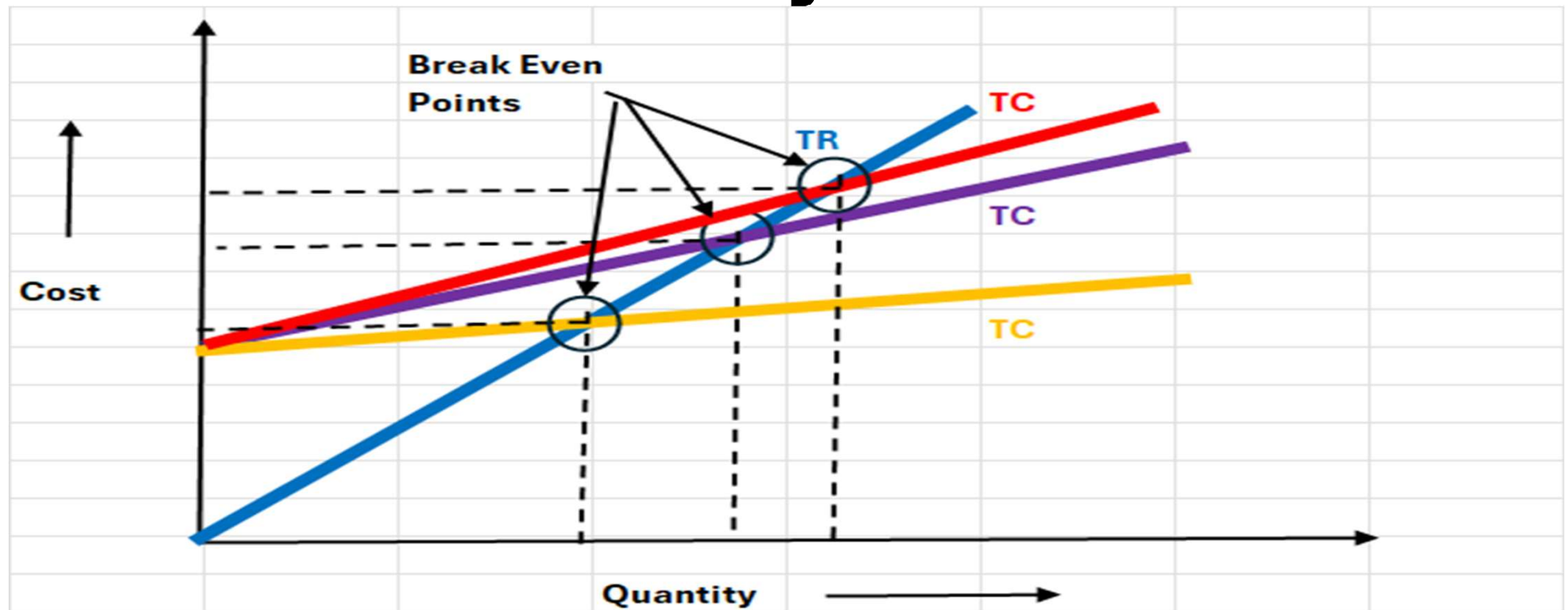
Capacity planning and process analysis

- $TC = FC + Q \times v$
- $TR = R \times Q$
- $P = TR - TC = R \times Q - (FC + Q \times v) = Q(R - v) - FC$
- $Q = (P + FC)/(R - v)$
- $Q_{BEP} = FC/(R - v)$, when $P = 0$

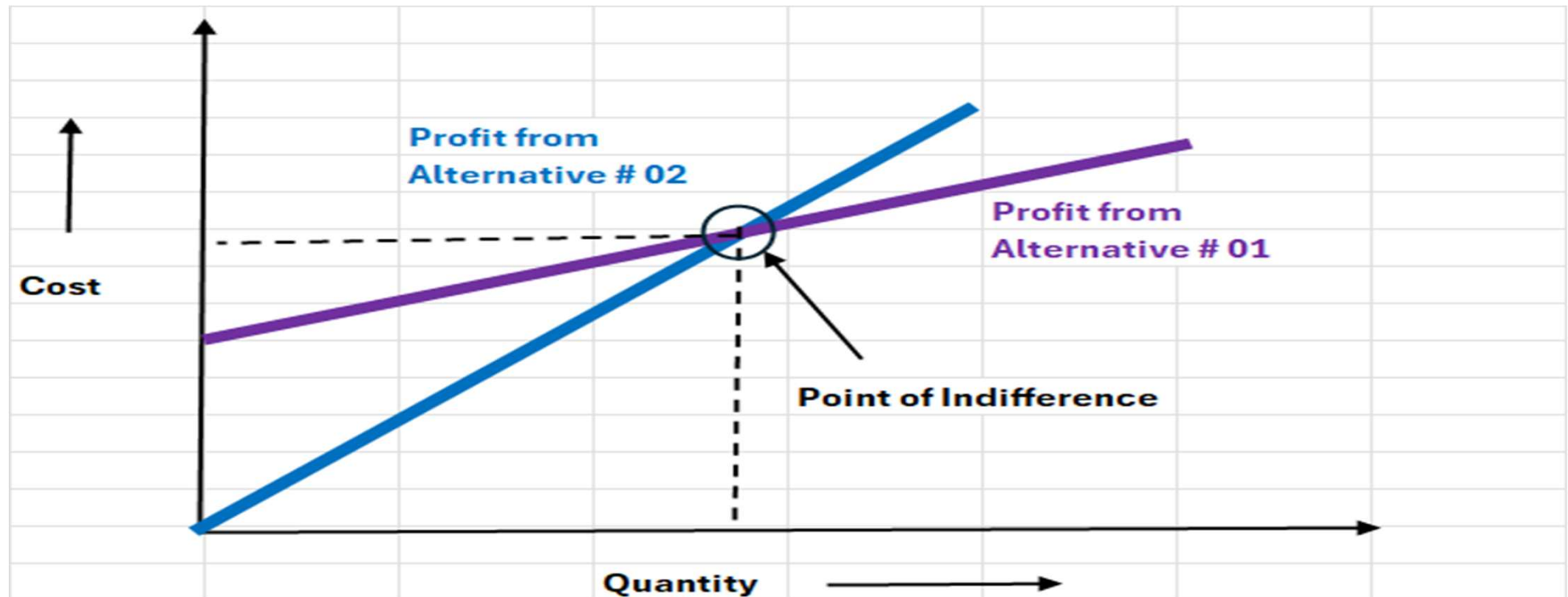
Capacity planning and process analysis



Capacity planning and process analysis



Capacity planning and process analysis



Capacity planning and process analysis

(Example # 17)

- The owner of DoAA Canteen at IIT Kanpur wants to find the business strategy for selling lunch *thalis*, for which he requires setting up new kitchen with cooking equipment which would cost Rs. 60,000/-.
- Variable costs would be Rs. 50/- per *thali* which would sell for Rs. 125/- per plate.
 - ❖ How many *thalis* must be sold in order to break even?
 - ❖ What would the profit (loss) be if 1,000 *thalis* are sold per week?
 - ❖ How many pies must be sold to realize a profit of Rs. 4,000/-?
 - ❖ If 1,500 *thalis* can be sold, and the canteen owner wants to make a profit of Rs. 5,000/-, then what price should be charged per *thali*?

Capacity planning and process analysis

(Example # 17)

- Break-Even Point, $Q_{\text{BEP}} = FC/(R - v) = 60000/(125 - 50) = 800 \text{ thalis}$.
- For selling 1,000 *thalis* per week, $P = TR - TC = R \times Q - (FC + Q \times v)$
 $= Q(R - v) - FC = 1000 \times 125 - (60000 + 1000 \times 50) = 1,25,000 - 1,10,000 = \text{Rs. } 15,000/-$.
- For a profit of Rs. 4,000/-, the number of *thalis* sold is $Q = (P + FC)/(R - v) = (4000 + 60000)/(125 - 50) = 853.33 \approx 854 \text{ thalis}$.
- If 1,500 *thalis* can be sold, and the canteen owner wants to make a profit of Rs. 5,000/-, then $R = (P + FC)/Q + v = (5000 + 60000)/1500 + 50 = \text{Rs. } 93.33/- \text{ per thali}$.

Capacity planning and process analysis

(Example # 18)

- A manager of a small shoe manufacturer has the option of purchasing one, two, three or four machines.
- Variable cost is Rs. 200/- per unit and revenue is Rs. 800/- per unit.

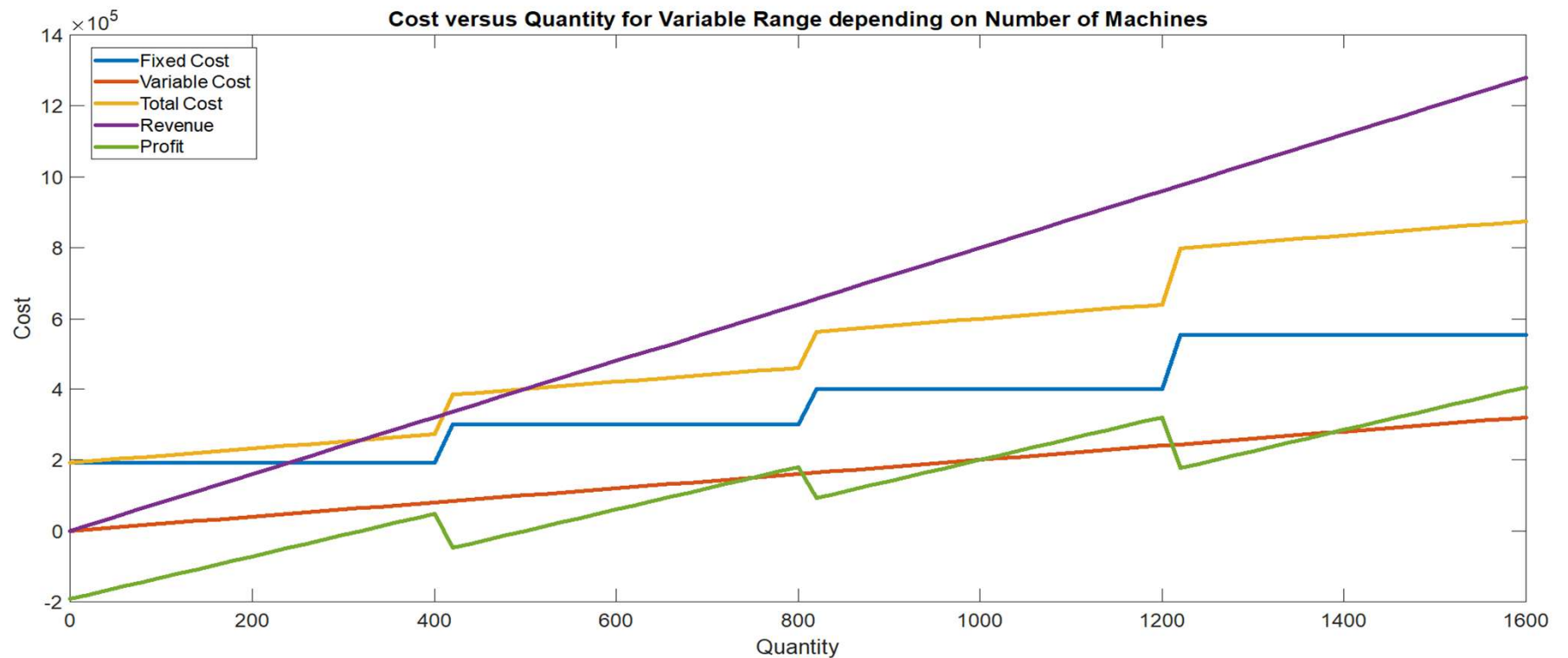
Number of Machines	Total Annual Fixed Costs (INR)	Corresponding Range of Output
1	1,92,000	0 to 400
2	3,00,000	401 to 800
3	4,00,000	801 to 1200
4	5,55,000	1201 to 1600

Capacity planning and process analysis

(Example # 18)

- The Break-Even Point for each range is as follows
 - ❖ $Q_{\text{BEP,Range \# 01}} = FC/(R - v) = 192000/(800 - 200) = 320$
 - ❖ $Q_{\text{BEP,Range \# 02}} = FC/(R - v) = 300000/(800 - 200) = 500$
 - ❖ $Q_{\text{BEP,Range \# 03}} = FC/(R - v) = 400000/(800 - 200) = 667$
(not in range hence not possible)
 - ❖ $Q_{\text{BEP,Range \# 04}} = FC/(R - v) = 555000/(800 - 200) = 925$
(not in range hence not possible)

Capacity planning and process analysis (Example # 18)



Capacity planning and process analysis

(Example # 19)

- A manager of a small shoe manufacturer has the option of purchasing one, two, three or four machines.

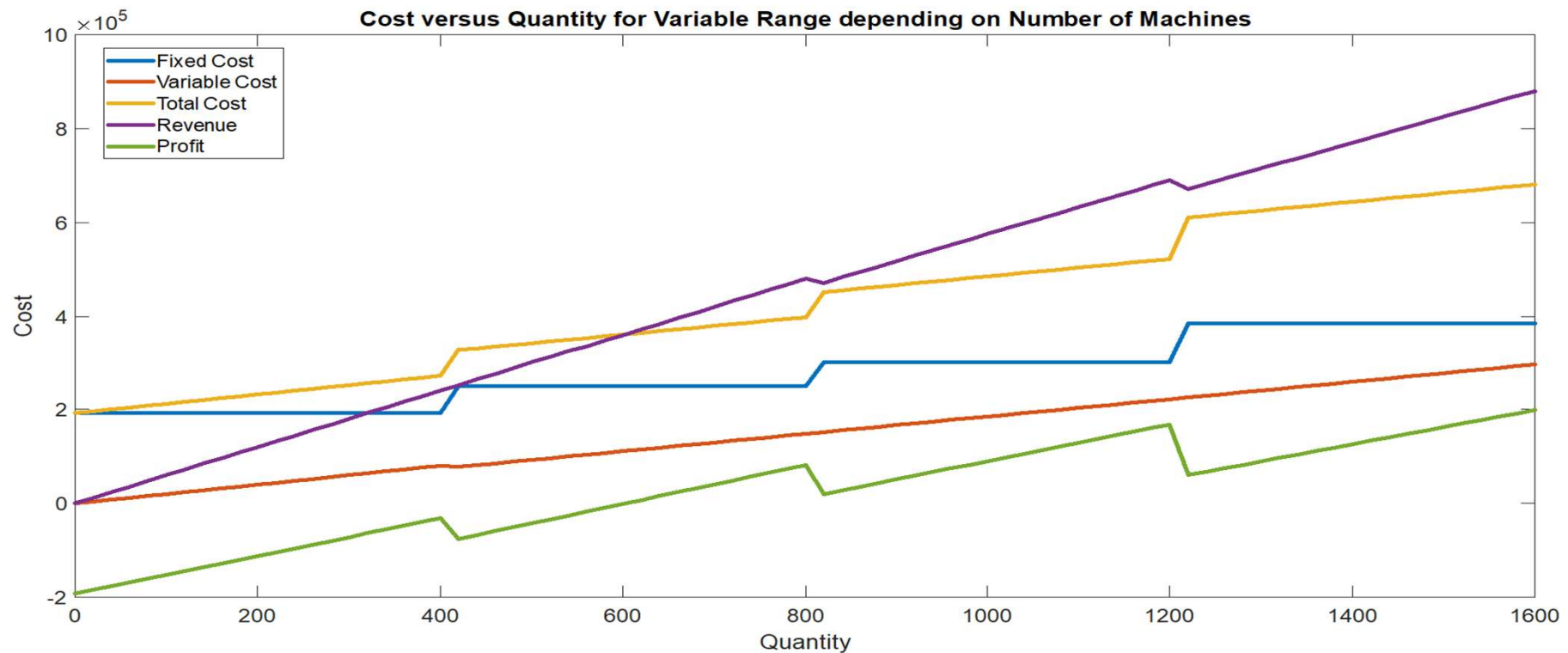
Number of Machines	Total Annual Fixed Costs (INR)	Corresponding Range of Output	Variable Cost per unit (INR)	Revenue per unit (INR)
1	1,92,000	0 to 400	200	600
2	2,50,000	401 to 800	185	600
3	3,00,000	801 to 1200	180	575
4	3,85,000	1201 to 1600	175	550

Capacity planning and process analysis

(Example # 19)

- The Break-Even Point for each range is as follows
 - ❖ $Q_{\text{BEP,Range \# 01}} = FC/(R - v) = 192000/(600 - 200) = 480$
(**not in range hence not possible**)
 - ❖ $Q_{\text{BEP,Range \# 02}} = FC/(R - v) = 250000/(600 - 185) = 603$
 - ❖ $Q_{\text{BEP,Range \# 03}} = FC/(R - v) = 300000/(575 - 180) = 760$
(**not in range hence not possible**)
 - ❖ $Q_{\text{BEP,Range \# 04}} = FC/(R - v) = 385000/(550 - 175) = 1027$ (**not in range hence not possible**)

Capacity planning and process analysis (Example # 19)



Capacity planning and process analysis

- Under financial analysis we have
 - ❖ Cash Flow Analysis
 - ❖ Net Present Value (NPV) Analysis
 - ❖ Internal Rate of Return (IRR)

Capacity planning and process analysis

- A : Total amount
- P : Principal amount
- r_{SI} : Simple Interest (SI)
- $r_{CI,m}$: Compound Interest (CI), with m compounding per year
- $r_{CI,\infty}$: Continuous Compounding Interest (CCI)
- n : Number of years

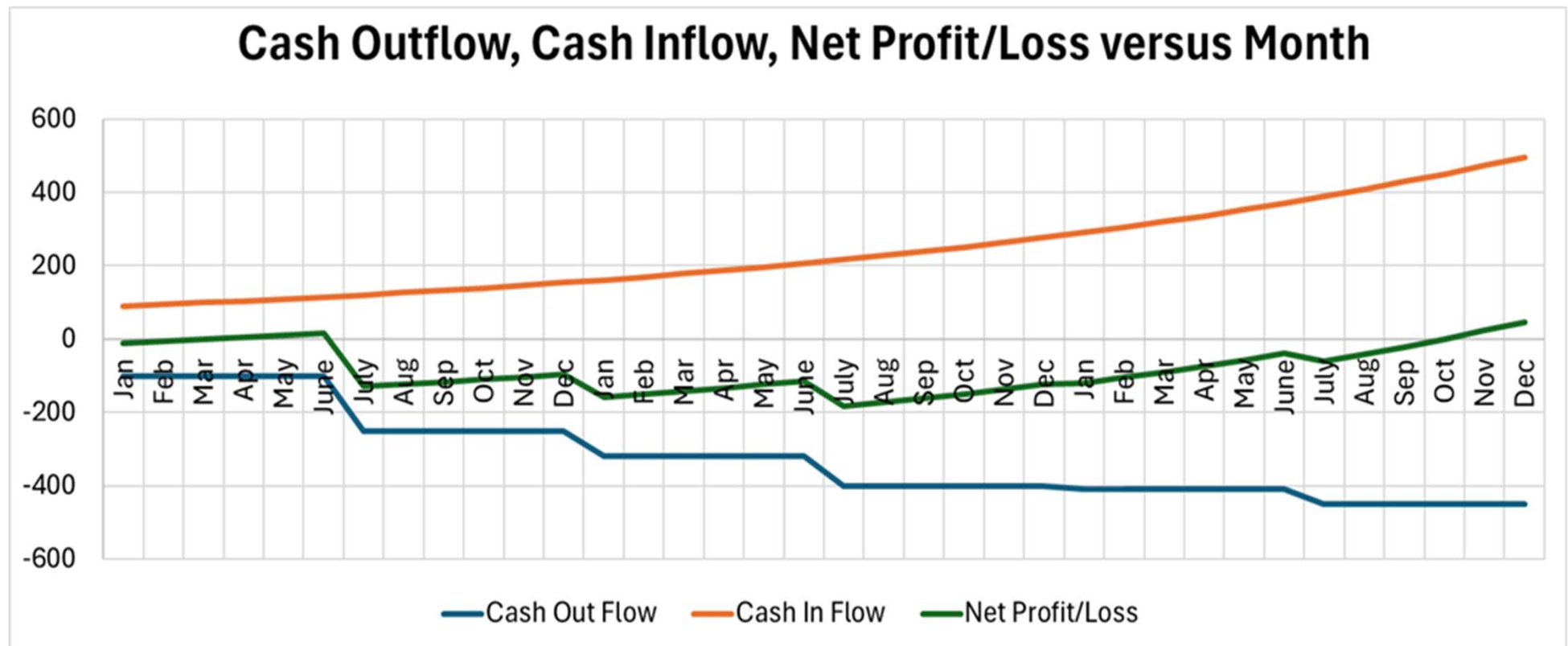
Capacity planning and process analysis

(Example # 20)

- $A = P(1 + r_{SI})^n$.
- $A = P(1 + r_{CI,m}/m)^{n \times m}$.
- $A = P \times \exp(r_{CCI,\infty} \times n)$.
- Relationship between CI with m_1 and m_2 respective number of times of compounding: $P(1 + r_{CI,m_1}/m_1)^{n \times m_1} = P(1 + r_{CI,m_2}/m_2)^{n \times m_2} = P \times \exp(r_{CI,\infty} \times n)$.
- $A = 100$ Euros, $r_{CI,\infty} = 4.229\%$ per annum, $n = 3$ months, then $100 \times \exp\{0.04229 \times (3/12)\} = 100 \times (1 + r_{CI,2}/2)^{(3/12) \times 2}$, hence $r_{CI,2} = 4.274\%$ per annum.

Capacity planning and process analysis

(Cash flow Analysis: Example # 21)



Capacity planning and process analysis (NPV: Example # 22)

- We are given the project outflow, inflow, CI % pa and time values are follows. Find NPV.

Month	Inflow (INR)	Outflow (INR)	CCI % pa	Month	Inflow (INR)	Outflow (INR)	CCI % pa
Jan	0	120	2.5	July	425	120	2.4
Feb	0	130	2.2	Aug	0	100	2.4
Mar	0	135	2.1	Sep	545	100	2.3
Apr	230	135	2.2	Oct	320	100	2.2
May	0	145	2.0	Nov	500	325	2.5
June	0	145	2.0	Dec	510	0	2.0

Capacity planning and process analysis (NPV: Example # 22)

- NPV = NPV of all Outflows + NPV of all Inflows
- NPV of all Outflows = $-[120 \times \exp\{-(2.5/100) \times (1/12)\} + 130 \times \exp\{-(2.2/100) \times (2/12)\} + \dots + 325 \times \exp\{-(2.5/100) \times (11/12)\} + 0 \times \exp\{-(2.0/100) \times (12/12)\}] = 1535.79$ Crores
- NPV of all Inflows = $+[0 \times \exp\{-(2.5/100) \times (1/12)\} + 0 \times \exp\{-(2.2/100) \times (2/12)\} + \dots + 500 \times \exp\{-(2.5/100) \times (11/12)\} + 510 \times \exp\{-(2.0/100) \times (12/12)\}] = 2485.85$ Crores
- NPV = $1535.79 + 2485.85 = 950.0609$ Crores

Capacity planning and process analysis

(IRR: Example # 23)

- We are given the project outflow, inflow, and time values are as follows. Find IRR.

Month	Outflow (INR)	Inflow (INR)	Month	Outflow (INR)	Inflow (INR)
Jan	190	100	July	100	155
Feb	180	110	Aug	90	150
Mar	165	115	Sep	90	100
Apr	150	125	Oct	75	95
May	120	130	Nov	60	90
June	120	145	Dec	50	80

Capacity planning and process analysis

(IRR: Example # 23)

- $NPV = 0$
- NPV of all Outflows = NPV of all Inflows
- NPV of all Outflows = $-[190 \times \exp\{-(r_{IRR}/100) \times (1/12)\} + 180 \times \exp\{-(r_{IRR}/100) \times (2/12)\} + \dots + 60 \times \exp\{-(r_{IRR}/100) \times (11/12)\} + 50 \times \exp\{-(r_{IRR}/100) \times (12/12)\}]$
- NPV of all Inflows = $+ [100 \times \exp\{-(r_{IRR}/100) \times (1/12)\} + 110 \times \exp\{-(r_{IRR}/100) \times (2/12)\} + \dots + 90 \times \exp\{-(r_{IRR}/100) \times (11/12)\} + 80 \times \exp\{-(r_{IRR}/100) \times (12/12)\}]$
- $r_{IRR} \approx 3.0260\%$ such that NPV of all Outflows = 1380 Crores and NPV of all Inflows = 1380 Crores

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Assumptions**)

- 1) Demand rate is constant and deterministic.
- 2) Order quantity need not be an integral number of units, and there are no minimum or maximum restrictions on its size.
- 3) Unit variable cost does not depend on the replenishment quantity; in particular, there are no discounts in either the unit purchase cost or the unit transportation cost.
- 4) Cost factors do not change appreciably with time; in particular, inflation is at a low level.

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Assumptions**)

- 5) Item is treated entirely independently of other items; that is, benefits from joint review or replenishment do not exist or are simply ignored.
- 6) Replenishment lead time is of zero duration; extension to a deterministic, nonzero replenishment lead time creates no problem.
- 7) Shortages are allowed.
- 8) Entire order quantity is delivered at the same time.
- 9) Planning horizon is very long, i.e., we assume that all the parameters will continue at the same values for a long time.

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Assumptions**)

- Considering
 - ❖ Shorting is not allowed, i.e., # 03 is not considered
 - ❖ Change in production rate and/or workforce is not allowed, i.e., # 05 is not considered
 - ❖ Replenishment lead time is zero, i.e., # 06 is not considered
- We need to find the optimal quantity to order (**Economic Order Quantity (EOQ)**) in order to find the optimal **Total Relevant Cost (TRC)** per unit time

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Different types**)

- 1) Basic production or purchase costs
- 2) Inventory carrying costs
- 3) Costs of insufficient capacity in the short run
- 4) Control system costs
- 5) Costs of changing work force sizes and production rates

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Calculations**)

- Q = Replenishment order quantity (in **units**)
- D = Demand rate of the item (in **units per year**)
- $T = Q/D$ = Time between placement (and receipt) of replenishment orders (in **years**)
- $N = D/Q$ = Number of replenishment **number per year**
- A = Fixed cost component with each replenishment (independent of magnitude of replenishment) (in **Rs.**)
- v = Variable cost per item (in **Rs. per unit**)
- r = Carrying cost/cost of having Re. 1/- of the item tied up in the inventory for a unit time interval (normally 1 year) (in **Rs. per Rs. per unit time**)
- p = Percentage (%) change in order quantity with respect to Q^* (**EOQ**)

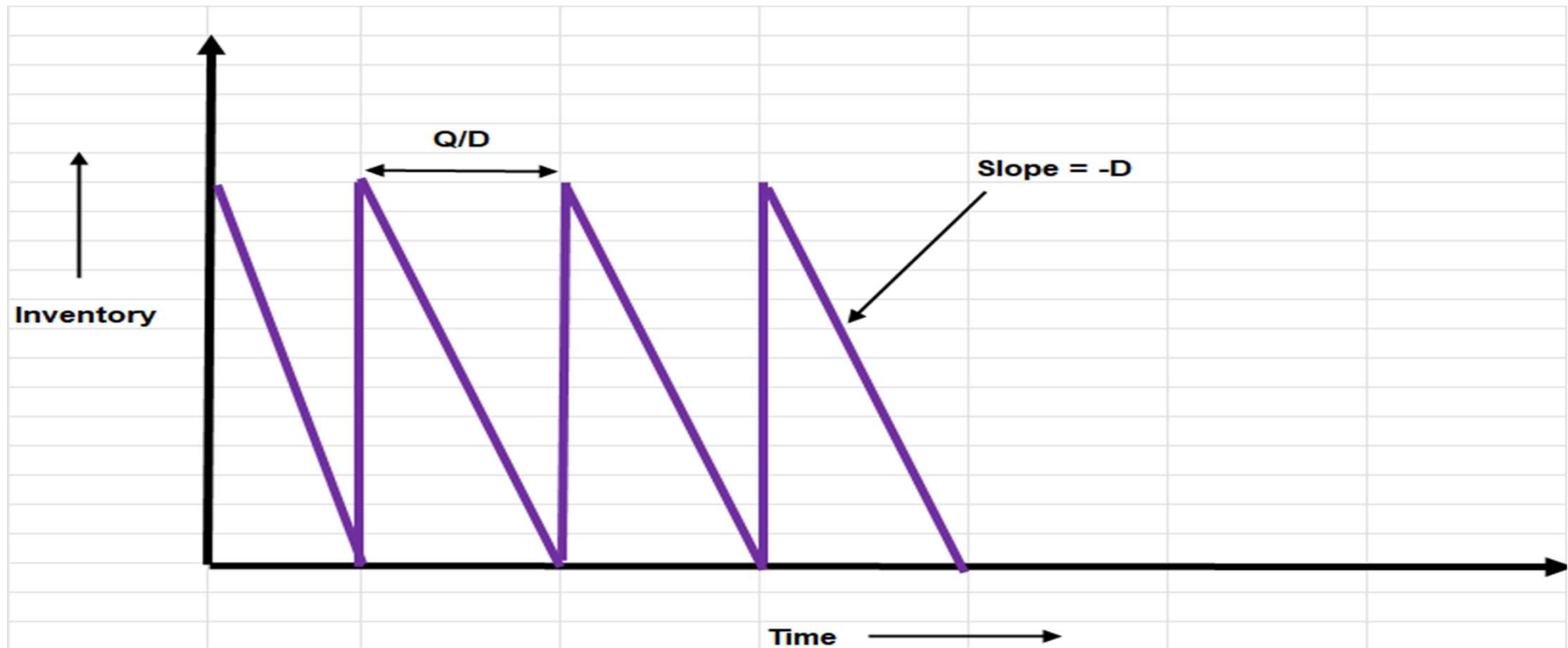
Economic Order Quantity (EOQ)/ Total Relevant Costs (**Calculations**)

- **Replenishment cost** Rs. per year = $(A + Q \times v) \times (D/Q) = (AD/Q + Dv)$, we assume v is **independent** of Q
- **Cost of carrying inventory** Rs. per year = $Qvr/2$
- **Total Relevant Costs (TRC)** Rs. per year i.e., $TRC(Q) = AD/Q + Dv + Qvr/2 = A/T + Dv + DTvr/2$
- Minimizing $TRC(Q)$, gives us $Q^* = \text{EOQ} = \sqrt{(2AD/vr)}$ in **units**
- $TRC(Q^*) = AD/\sqrt{(2AD/vr)} + Dv + \sqrt{(2AD/vr)} \times vr/2 = \sqrt{(2ADvr)} + Dv$ in **Rs. per year**

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Calculations**)

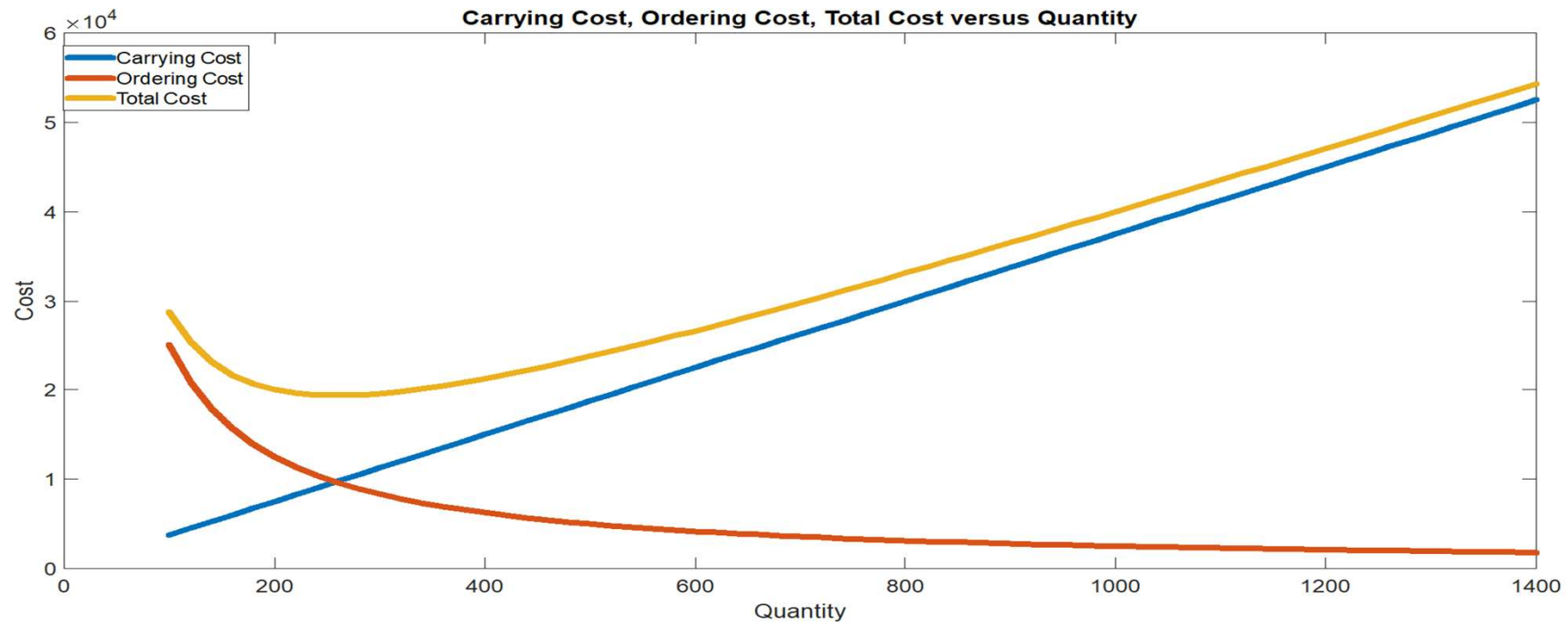
- **Time between order**, $T(Q) = (Q/D)$, thus $T(Q^*) = T^* = \{\sqrt{(2AD/vr)}\} \times (1/D) = \sqrt{(2A/Dvr)}$ (in years)
- $TRC(T) = AD/Q + Dv + Qvr/2 = A/T + DTvr/2 + Dv$, thus $TRC(T^*) = A/T^* + DT^*vr/2 + Dv$ (in Rs. per year)
- **Inventory turnover ratio (TR)** (annual demand rate divided by the average inventory level) and this gives the number of times the average inventory is sold or turned over some given time period, $TR(Q) = D/(Q/2)$, thus $TR(Q^*) = \sqrt{(2Dvr/A)}$ (in number)
- If $Q = (1 \pm p)Q^*$, then **Percentage Cost Penalty (PCP)** can be found by replacing $Q = (1 \pm p)Q^*$ in place of Q^*

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Order Quantity and Inventory versus Time**)



Economic Order Quantity (EOQ)/ Total Relevant Costs (CC, OC, TC, EOQ: Example # 24)

- $v=15$; $r=5$; $A=2500$; $D=1000$



Economic Order Quantity (EOQ)/ Total Relevant Costs (**Example # 25**)

- D : Demand for an item used in production is relatively stable over time and is 2,000 (in units per year)
- A : Fixed cost per replenishment is Rs. 1,500/- (in Rs.)
- v : Unit variable cost is Rs. 20/- (in Rs. per unit)
- r : Carrying cost/cost is 30 per year (in per year)
- p = Percentage change in order quantity Q with respect to Q^* is (+25% or -25%)

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Example # 25**)

- **EOQ** = **Q^*** = $\sqrt{(2AD/vr)}$ =
 $\sqrt{(2 \times 1500 \times 2000 / 20 \times 30)} = 100$ in units
- **Total Relevant Costs for Q^* = $TRC(Q^*)$** =
 $\sqrt{(2ADvr)}$ + Dv =
 $\sqrt{(2 \times 1500 \times 2000 \times 20 \times 30)} + 20 \times 2000$ =
 $60000 + 40000 = 100000$ in Rs.

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Example # 25**)

- **Time between order for Q^* = T^* =**
 $\sqrt{(2A/Dvr)} = \sqrt{(2 \times 1500 / 2000 \times 20 \times 30)} =$
 0.05 (in years) $= 0.05 \times 12 = 0.6$ (in months)
- **Inventory turnover ratio for Q^* = $TR(Q^*)$**
 $= \sqrt{(2Dvr/A)} = TR(Q^*) =$
 $\sqrt{(2 \times 2000 \times 20 \times 30 / 1500)} = 40$ in numbers

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Example # 25**)

- **Total Relevant Costs for $Q=Q^*(1+0.25)$** =
 $(AD/Q + Dv) + Qvr/2 = (1500 \times 2000 / 125 + 2000 \times 20) + 125 \times 20 \times 30 / 2 = 101500$ in Rs.
- **Total Relevant Costs for $Q=Q^*(1-0.25)$** =
 $(AD/Q + Dv) + Qvr/2 = (1500 \times 2000 / 75 + 2000 \times 20) + 75 \times 20 \times 30 / 2 = 102500$ in Rs.

Economic Order Quantity (EOQ)/ Total Relevant Costs (**Example # 25**)



Operations Management

- Introduction to production management
- Capacity planning and process analysis
- **Network Analysis**
- Operations Research

Network Analysis

Analysis of Performance

- To study, understand and analyze any process and study its performance one needs to draw the **process flow diagram** in detail.
- **Process analysis**: Is a rigorous framework for understanding the detailed operations of a business. It determines how many flow units can be processed per unit of time (the **process capacity**) and how busy the resources of the process are (**utilization**).
- While analyzing the process one is interested to study the following
 - ❖ Activities
 - ❖ Buffer Inventory
 - ❖ In Process Inventory
 - ❖ Resources
 - ❖ Utilization of resources
 - ❖ Flow of inventory from input stage to output stage
 - ❖ Down stream inventory
 - ❖ Up stream inventory

Analysis of Performance

(Example # 26)

- Consider Subway inside IIT Kanpur. The eatery has the following sections (designated areas)
 - ❖ Sitting place (A)
 - ❖ Fridge/Coffee (B)
 - ❖ Non-Veg (C)
 - ❖ Veg (D)
 - ❖ Bread heater (E)
 - ❖ Kitchen/Storage (F)
 - ❖ Entry (G)

Analysis of Performance

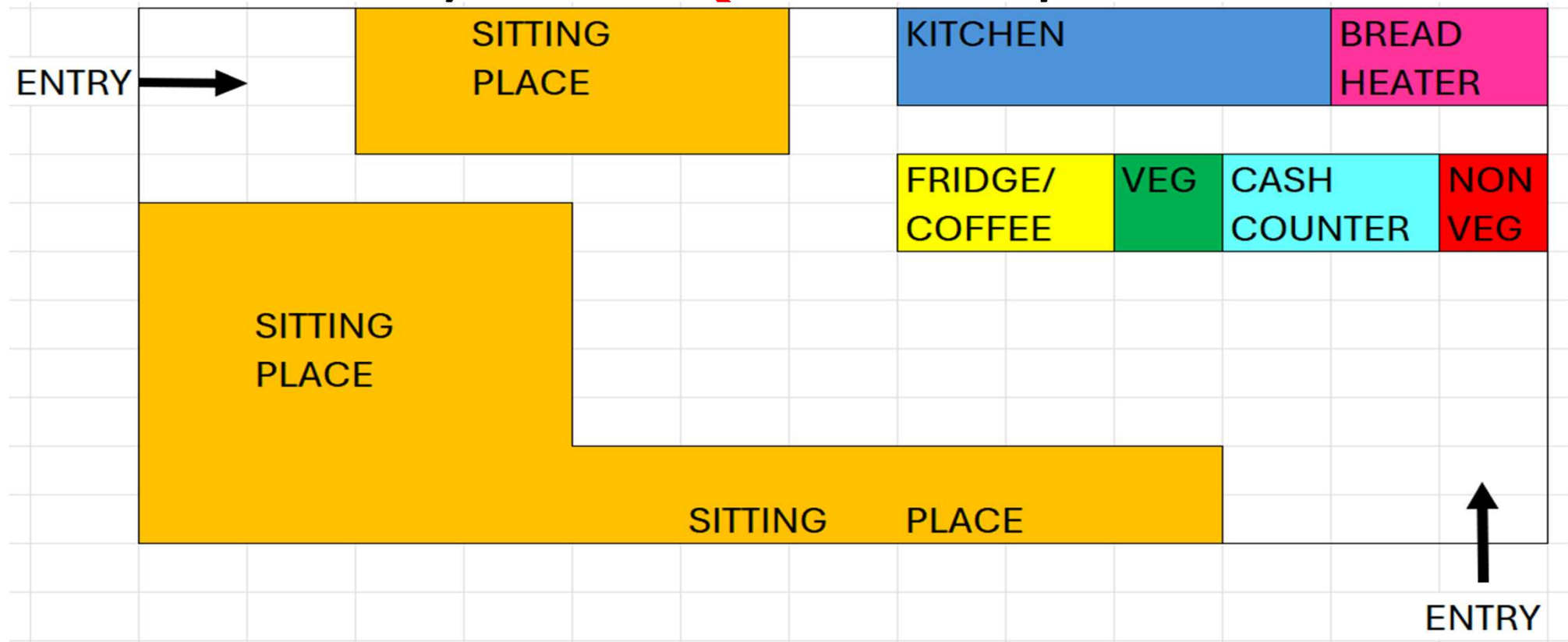
(Example # 26)

- The **activity relationship** chart is given below

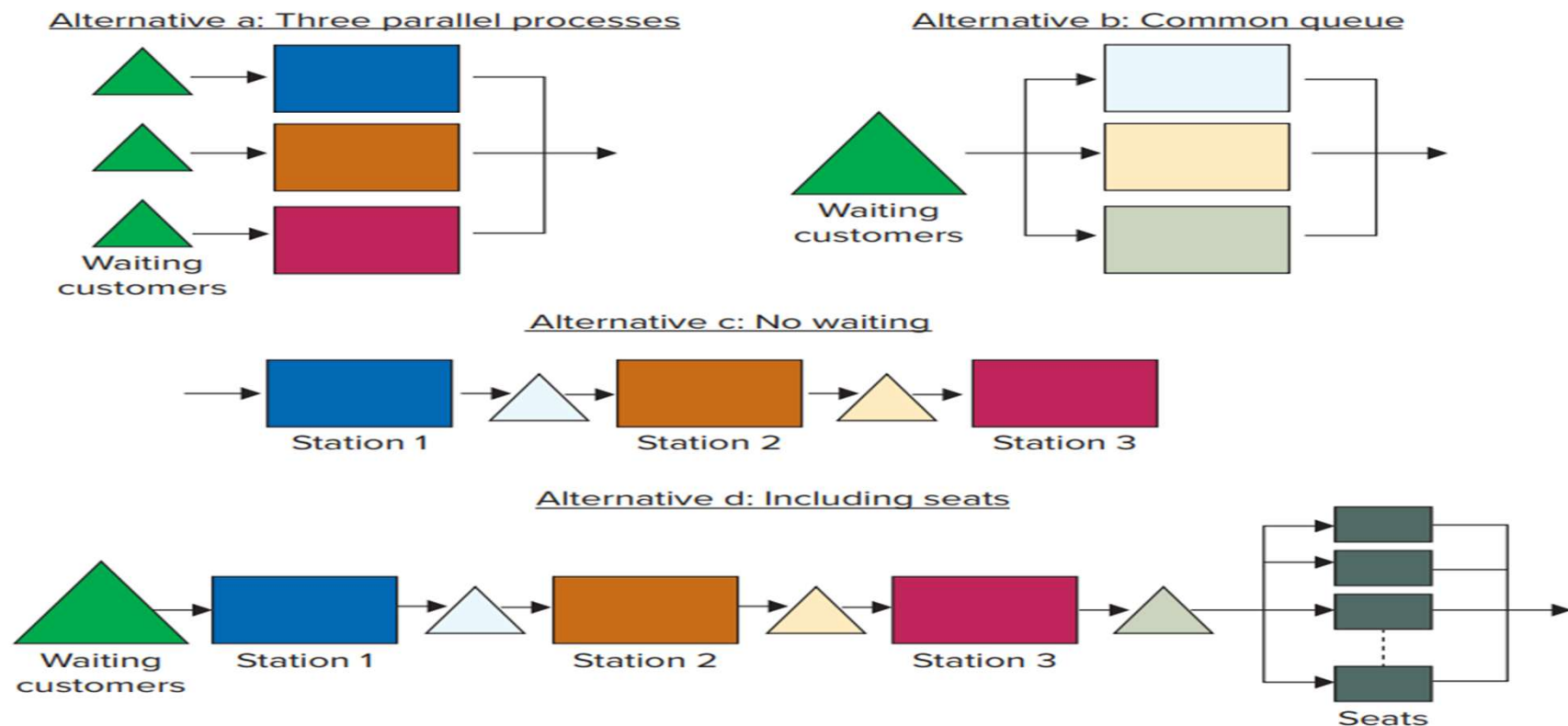
		A	B	C	D	E	F	G
Sitting Space	A	-	++++	++	++	+	+	+
Fridge/Coffee	B	++++	-	++	+++	++	+	+
Non-Veg	C	++	+++	-	+	+	+	+
Veg	D	++++	+++	+	-	+	+	+
Bread Storage/Heater	E	+	+++	+	+++	-	+++	+
Kitchen/Storage	F	+	+++	+++	+++	++++	-	+
Entry	G	+	+++	+	+	+	+	-

Analysis of Performance

(Example # 26)



Analysis of Performance (Diagrams)



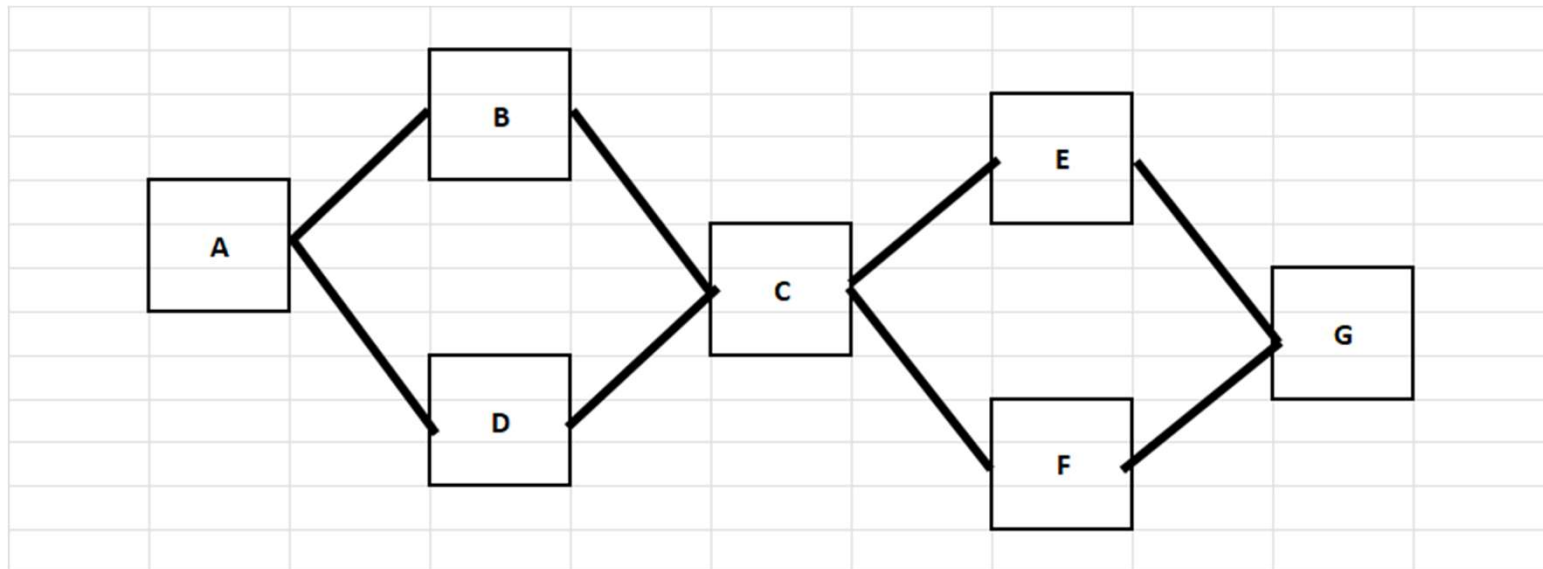
Analysis of Performance

(Example # 27)

- Consider a machining shop floor the **from to chart** with distances (in meters) are show below

To 		A	B	C	D	E	F	G
From 								
Drilling	A	-	20	60	25	80	90	110
Grinding	B	20	-	30	35	30	45	65
CNC	C	60	30	-	30	15	25	40
Lathe	D	25	35	30	-	40	35	40
Sub-part # 01 Assembly	E	80	30	15	40	-	15	25
Sub-part # 02 Assembly	F	90	45	25	35	15	-	20
Final Assembly	G	110	65	40	40	25	20	-

Analysis of Performance (**Example # 27**)



Analysis of Performance (**Example # 27**)

- Processing time at each machine
- Change over time at each machine
- Time to travel from one machine to the next in the sequence
- Individual inventory levels at each machine
- Flow process inventory between machines in the flow process

Analysis of Performance (CPM and PERT)

- **Cost estimation** is needed and it can be deterministic as well as stochastic
- **Time estimation** is needed and it can be deterministic as well as stochastic
- Typically, there are two ways to determine activity durations:
 - ❖ **Deterministic**: Assumes that sufficient knowledge is available to develop a reasonably accurate time estimate for project activities, **Critical Path Method (CPM)**
 - ❖ **Stochastic**: Assumes that knowledge is in sufficient to develop any reasonably accurate time estimate for project activities, **Project Evaluation Review Technique (PERT)**

Analysis of Performance (CPM and PERT)

- There are three terms connected to scheduling that we need to define and they are
 - ❖ **Activity** – which is a set of work items requiring resources to be executed
 - ❖ **Event** – which is a point in time when an activity starts or finishes
 - ❖ **Milestone** – which is a planned observable event connected to a defined result
- There are two ways to represent a schedule:
 - ❖ **GANTT** chart (bar chart)
 - ❖ **Network**

Analysis of Performance (GANTT Chart)

